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Prepositioning Network Design for Disaster Relief Purposes under Spatially Correlated Demand Uncertainty

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Abstract Prepositioning disaster relief network is one of the most essential elements in natural disaster management. However, designing a proper network is challenging due to limited information and, more importantly, the spatially correlated demand uncertainty that exists among affected areas. We consider a network design problem for disaster relief purposes, where spatial demand correlations exist and demand information is limited, i.e., only the mean demand and covariance matrix are known. Note that the covariance matrix can explicitly capture the spatially correlated demand among areas. We formulate this problem as a mixed-integer two-stage distributionally robust location-inventory model, which is generally NP-hard and computationally intractable. The model is further reformulated as a mixed-integer conic problem based on copositive cones, and it is tractable with positive semidefinite relaxation. To accelerate the problem-solving process, we design an interpretable branching-and-pricing heuristic with a warm start. Both case study and simulation results demonstrate that explicitly modelling spatial demand correlation is necessary. Compared to a sample-average-approximation model and a distributionally robust model without demand correlation information, our model can decrease unmet demand by approximately 18% and 48%, respectively, without increasing total costs.

Keywords Disaster relief network design · Distributionally robust optimization · Copositive programming

1 Introduction

Natural disasters and catastrophes have always existed in the world. Although our society has developed advanced response mechanisms to disasters and modern disaster monitoring systems, losses can still be immense. Every year, approximately 460 reported dramatic natural disasters, such as earthquakes, volcanoes, landslides, famines, droughts, hurricanes, etc., cause approximately 60,000 deaths worldwide, responsible for 0.1% of global deaths, and lead to 0.05% to 5% economic losses in relation to gross domestic product by different countries (Ritchie and Roser (2014)). For example, the 2008 Wenchuan earthquake, a magnitude 8.0 earthquake in China, entailed life losses of over 69,000 people, injuries of over 370,000 people, and

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missings person accounts of over 18,000 people. In addition to devastating impacts in terms of the loss of human life, the Wenchuan earthquake caused severe destruction with economic costs. AIR Worldwide, a catastrophe modelling firm reported that the estimated direct total economic cost due to the Wenchuan earthquake exceeded US$20 billion.

Given the negative influence that natural disasters can impose on mankind and society, experts from different areas have been striving to alleviate suffering in terms of both lives and economics through various aspects. To better understand the disaster process and harness natural disasters, Gupta et al. (2016) generally categorized disaster time phases as three phases: before the disaster, during the disaster, and after the disaster. However, many types of natural disasters only possess two phases: before (preparedness phase) and after (response phase), such as earthquakes. This is reasonable since the devastating strike of an earthquake usually lasts only minutes, during which efficient emergency service and humanitarian support cannot effectively operate. Specifically, we restrict the natural disaster we consider to earthquakes.

Disaster relief network design is a classic but essential approach to encounter the damage caused by earthquakes. Earlier literature (Tufekci and Wallace (1998)) in the field of disaster management indicated that regarding the preparedness phase and the response phase as two separate parts would cause sub-optimality. To design a reliable relief network, a two-stage location-inventory model taking into account both the preparedness phase and response phase is necessary. Specifically, decision makers are required to determine where to establish warehouses from a set of predetermined potential locations and then preposition appropriate relief materials correspondingly in the preparedness phase. The objective during the preparedness phase is to minimize the network setup cost plus expected future economic loss due to relief material shortages. In the response phase, after an earthquake occurs and relief material demand realizes, the decision maker has to address a resource allocation problem, that is how to allocate the deployed relief materials to fulfill the demand as much as possible. Note that each warehouse can serve several different demand areas within its service region, and a demand area can be served by several different warehouses. In disaster management problems, the high demand fulfilment can greatly alleviate loss in terms of lives and economics. Therefore, demand fulfilment is one of the most critical performance measures and should be taken into account in the network design.

The main challenge of designing a reliable relief network lies in the demand uncertainty posed by the earthquake. Typically, demand independence among different areas is assumed to make optimization models more tractable. However, due to the inherent nature of earthquakes, demand among areas is spatially positively correlated. This means that the higher the realized demand from one area is, the higher the potential higher demand from another area will be. A model without explicitly considering this unique feature instead of assuming spatial independence would yield an overly "optimistic" network. Our meaning of optimistic is meant to emphasize that the resulting network possibly holds fewer relief materials. As a result, the shortage of relief materials after an earthquake occurrence would be massive.

To tackle the correlated uncertainty of demand, two main approaches are traditionally being utilized: stochastic programming and distributionally robust optimization. In stochastic programming, the correlated uncertain demand is represented as random variables with a known joint probability distribution. However, as disasters are rarely occuring events, the joint distribution of demand is hard to predict from historical data due to the scarcity of data points. Thus, stochastic programming is not suitable for our scenario due to the scarcity of historical data. Instead, we adopt a distributionally robust optimization (DRO) technique to address the uncertainty of demand. The aim of DRO is to minimize the worst-case expected total cost over a set of feasible demand distributions. Specifically, we adopt moments-based DRO, in which the set of distributions is characterized by the moments of demand, e.g., the mean value and covariance matrix. The motivation of adopting a covariance matrix rather than only variance is that demand is highly spatially correlated in an earthquake situation, as mentioned above. Mathematically, non-diagonal elements of the covariance matrix could describe such relationships correctly. Thus, developing a model explicitly incorporating the covariance matrix of demand offers the possibility of capturing spatially correlated uncertainty of demand among areas.

In this paper, we address a prepositioning network design problem for disaster relief purposes under spatially correlated demand uncertainty. First, we develop a model, explicitly considering the correlation among demand, to jointly address decisions in the preparedness and response phases. To our best of knowledge, we are the first to incorporate demand correlation into the Location-Inventory model with DRO technique and apply it to disaster management. Then, we reformulate the model based on copositive programming and obtain an exact mixed-integer formulation. Finally, we demonstrate the effectiveness of explicitly modelling correlations by conducting simulations on both synthetic datasets and a real-world earthquake case. Therefore, our main contributions can be summarized as follows:

1. We introduce spatially correlated demand uncertainties to disaster relief network design problems by specifying the demand covariance matrix. This problem is described as a mixed-integer two-stage distributionally robust optimization model (see Model (DRND)).
2. We mathematically reformulate Model (DRND) as a mixed-integer copositive programming problem (see Proposition 4), solving which generates the well-designed network.
3. We propose an interpretable branching method with a warm start to accelerate the branch-and-pricing solving process, thereby cutting the computational time in half (see Algorithm 1).
4. Numerical experimental results based on both synthetic data and real data demonstrate that the model with demand correlation can provide a more reliable relief network.

Additionally, we summarize assumptions that we follow in developing models.

1. Warehouses do not suffer from disruptions caused by disasters, as well as relief materials.
2. Roads through which relief materials are delivered do not become inaccessible and have unlimited transportation capacity.

The remainder of this paper is organized as follows. We first present a literature review of disaster relief network design with demand uncertainties and copositive programming, a math tool we utilized greatly, in Section 2. We formally propose our model in Section 3, where we also propose the reformulated model and an algorithm to accelerate the solution process. In Section 5, we conduct numerous numerical studies to illustrate the advantages of the proposed models. In Section 6, several extensions are proposed to demonstrate our model's applicability to more complex situations. Finally, in Section 7, we conclude our study.

Notations. Throughout this paper, we use lowercase ***bold*** and UPPERCASE ***bold*** characters to denote vectors and matrices, respectively. Corresponding normal characters denote component-wise elements. Variables with tilde symbols, such as $\tilde{\boldsymbol{d}}$, represent random variables. We use $\boldsymbol{A} \succeq \boldsymbol{0}$ to indicate that matrix $\boldsymbol{A}$ is positive semidefinite. $\boldsymbol{A} \succeq_{co} \boldsymbol{0}$ and $\boldsymbol{A} \succeq_{cp} \boldsymbol{0}$ require that $\boldsymbol{A}$ belongs to a copositive or completely positive cone with proper dimensions. $\mathbb{R}^n_+$ denotes an n-dimensional nonnegative real space, and $[n]$ represents the running set $\{1, 2, 3, \ldots, n\}$. $\mathbb{E}_{\mathbb{P}}[\cdot]$ is the expectation with respect to distibution $\mathbb{P}$. We use $\tilde{\boldsymbol{d}} \sim \mathbb{P}$ to indicate that the random variable $\tilde{\boldsymbol{d}}$ follows distribution $\mathbb{P}$. And $\mathcal{P}(\mathbb{R}^n_+)$ defines a distribution family consisting of all measurable distributions on an n-dimensional nonnegative support set.

2 Literature Review

2.1 Disaster Relief Network Design under Demand Uncertainty

The study of disaster relief network design started in 1958. It was first proposed by Baumol and Wolfe (1958), which describes a heuristic for a warehouse location model under a single-sourcing setting. Thereafter, a large number of variants that considered uncertainties were proposed to address more complex situations. Since a broad strand of literature on disaster relief network design exists and we review only related and representative works here, interested readers are encouraged to read Behl and Dutta (2019) for a comprehensive review.

A classic approach to model the demand uncertainty is by assuming the probability distributions or some senarios are known to the decision maker. For example, Tamura et al. (2000) constructs a probability tree to model disaster risks. Dodo et al. (2007) uses a set of 47 earthquake scenarios to help make the infrastructure development decisions. Zhang et al. (2019) considers secondary disasters with the conditional probability based scenario tree. Given the computational tractability of senario-based models, many disaster features have been embedded, such as helicopter operations (Barbarosoğlu et al. (2002)), the loss of part of the prepositioned items (Pradhananga et al. (2016)), the occurrence of disruptions of roads (Vitoriano et al. (2011)), and disaster propagation processes (Klibi et al. (2018)). Hoyos et al. (2015) conducts a literature survey on Operations Research (OR) models with stochastic components in disaster management, to which interested readers are referred.

All the models in the above discussion can be generalized as stochastic programming problems. However, this kind of stochastic programming problem is widely accused of assuming a known but possibly wrong distribution, especially in the field of disaster management. In contrast to stochastic models, another widely used tool is robust optimization in which only the support set information is known. For example, Ben-Tal et al. (2011) adopts a min–max criterion to dynamically assign emergency materials in time-dependent uncertainty scenarios. Najafi et al. (2013) designs a robust earthquake response mechanism by assuming that information about the transportation network is easily accessible. Ni et al. (2018) proposes a min-max robust model to jointly choose locations and inventory levels. Rezaei-Malek et al. (2016) designs a robust disaster relief network with perishable commodities. Yahyaei and Bozorgi-Amiri (2019) considers relief logistics network design under interval uncertainty and the risk of facility disruption. While a robust disaster relief network can encounter extreme cases, it is costly and performs with lower satisfaction in terms of average metric in general.

Therefore, distributionally robust optimization (DRO) is adopted to mitigate both drawbacks. As an emerging modelling tool, DRO has been widely utilized to address operation problems with uncertainties. However, we found that studies that apply DRO to disaster relief network are quite limited, mostly because of the computational intractablity of DRO. For example, Liu et al. (2019) present a two-stage model to address emergency medical service station location and sizing problems. Although they also consider the moment information, they define their ambiguity set with an ellipsoidal set proposed by Delage and Ye (2010), which simplifies the model to conic programming. Wang and Chen (2020) considers a blood supply network with disasters, utilizing distributionally robust optimization in which the ambiguity set is characterized by moments information. They proposes an approximate way to transform their model into a semidefinite program, but fail to give the exact reformulation. A very similar disaster relief network design problem is considered in Nakao et al. (2017). Instead of involving cross-moment information, they consider only marginal moment information, i.e., mean and variance. Moreover, they use an affine approximation to approximately solve the model. To the best of our knowledge, we are the first to explicitly consider cross-moment uncertainty of demand in a disaster relief network problem, and develop the exact reformulation.

2.2 Moment-based Distributionally Robust Optimization and Copositive Programming

Moment-based distributionally robust optimization is a robust formulation for stochastic programming, where the objective function is formulated with respect to the worst-case expected value over the choice of a distribution that is subject to an ambiguity set constrained by a known support set and moment information. The past decade witnessed explosive growth in research on this topic after Bertsimas and Popescu (1999) connected the moment problem and semidefinite optimization (see Mishra et al. (2012) and Bertsimas et al. (2019)). In practical problems, the support set should be restricted to a nonnegative orthant. For example, demand should always be nonnegative, and service time in clinics should be greater than zero. Natarajan et al. (2011) showed that, given the non-negative support set, mean value, and covariance matrix, computing the worst-case expected value of a problem with binary decision variables

and uncertainties in the objective function can be reformulated as a conic problem based on completely positive cones. This reformulation sparked extensive applications based on a completely positive cone or its dual cone, i.e., the copositive cone.

An n-dimensional copositive cone is defined as

$$\mathcal{CO}_n := \left\{ \boldsymbol{A} \in S_n \,\middle|\, \boldsymbol{v} \in \mathbb{R}_+^n, \boldsymbol{v}^T \boldsymbol{A} \boldsymbol{v} \geq 0 \right\}$$

where S_n denotes the cone of an $n \times n$ symmetric matrix. For $\boldsymbol{A} \in \mathcal{CO}_n$, we write $\boldsymbol{A} \succeq_{co} \boldsymbol{0}$. A completely positive cone is the dual cone of a copositive cone (Dickinson (2013)) and is defined as

$$\begin{aligned} \mathcal{CP}_n &:= \left\{ \boldsymbol{A} \in S_n \,\middle|\, \exists \boldsymbol{V} \in \mathbb{R}_+^{n \times m}, s.t. \boldsymbol{A} = \boldsymbol{V}\boldsymbol{V}^T \right\} \\ &:= \left\{ \boldsymbol{A} \in S_n \,\middle|\, \exists \boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_m \in \mathbb{R}_+^n, s.t. \boldsymbol{A} = \sum_{i=1}^{m} \boldsymbol{v}_i \boldsymbol{v}_i^T \right\} \end{aligned}$$

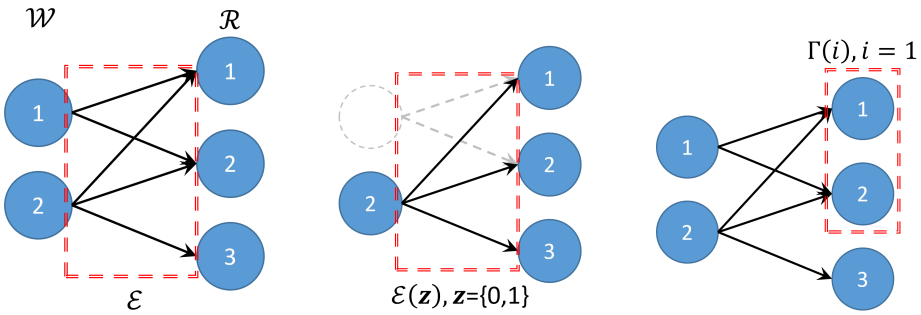
A conic programming problem where the cone is restricted to copositive cones or completely positive cones is called copositive programming or completely positive programming, respectively. Recently, copositive programming has attracted considerable attention in the optimization community as a powerful modelling technique. Many NP-hard problems have been reformulated via copositive programming or completely positive programming. For example, the well-known maximum stable set problem has been proven to be equivalent to a copositive program by modifying the feasible region of the theta-number problem from a semidefinite cone to a copositive cone (De Klerk and Pasechnik (2002)). In general, Burer (2009) concluded that any mixed-integer quadratic problems can be reformulated as a copositive programming problem. Additionally, Natarajan et al. (2011) introduced copositive programming into mixed 0-1 linear programs when coefficients of decision variables in the objective function themselves are random.

Although the original NP-hard problem could be reformulated as a copositive problem, the difficulty of optimization is not reduced. Indeed, checking whether a given matrix lies in a copositive cone has been observed to be NP-complete. However, such reformulations shift the difficulty of optimization to understanding the mathematical properties of copositive cones. Complete knowledge of the copositive cone could be used to solve many optimization problems more efficiently. Natarajan et al. (2011) proved that under mild conditions, completely positive programming problems could be solved exactly via semidefinite programming. A commonly used outer approximation for completely positive cones is the "doubly nonnegative cone", defined as $\{\boldsymbol{A} \,|\, \boldsymbol{A} \succeq \boldsymbol{0}, \boldsymbol{A} \geq \boldsymbol{0}\}$. Moreover, $\{\boldsymbol{A} \,|\, \boldsymbol{A} = \boldsymbol{A}_1 + \boldsymbol{A}_2, \boldsymbol{A}_1 \succeq \boldsymbol{0}, \boldsymbol{A}_2 \geq \boldsymbol{0}\}$ provides an inner approximation to copositive cones (Natarajan et al. (2011)). In our numerical study, we use this inner approximation to approximate copositive cones. Therefore, in what follows, the statement "solve copositive programming" actually means solving copositive programming with its relaxations.

In recent years, researchers have found that many operation management problems can be properly handled via copositive programming and its approximation techniques. For example, Natarajan et al. (2011) applied copositive programming to a project management problem to estimate the expected completion time and the persistence of each activity. Kong et al. (2013) solved a healthcare appointment-scheduling problem, where only the mean and covariance of service duration are known. They formulated the problem as a convex copositive optimization problem with a tractable semidefinite relaxation. Li et al. (2014) studied a sequencing problem with random costs. Utilizing copositive programming, Yan et al. (2018) designed a roving team deployment plan for Singapore Changi Airport. Kong et al. (2020) further incorporated patients' no-show behaviour into a healthcare appointment scheduling problem and reformulated the problem as a copositive program. We refer interested readers to Dickinson (2013) for the general analytic properties of copositive cones and to Li et al. (2014) for a comprehensive review of applications of copositive programming.

3 Disaster Relief Network Design Problem

Consider a Disaster Relief Network Design (DRND) problem with a set of potential supply nodes denoted by $\mathcal{W} = \{1, 2, \ldots, m\}$ and a set of demand nodes denoted by $\mathcal{R} = \{1, 2, \ldots, n\}$. The set of roads or links (or network structure) is denoted by $\mathcal{E}$ whose elements are pairs of supply nodes and demand nodes such as $\mathcal{E} := \{(1,1), (1,2), (2,1), (2,2), (2,3)\}$ (see Figure 1a), and its cardinality is $|\mathcal{E}| = r$. To rule out absurd cases, we assume $\mathcal{E}$ always defines a bipartite graph without isolated nodes. In the first stage, the decision maker designs the accessible network $\mathcal{E}(\boldsymbol{z})$ by selecting a subset of potential supply nodes to build warehouses and deploys an appropriate number of relief materials. Mathematically, we define the binary decision variable $\boldsymbol{z} = (z_1, z_2 \ldots, z_m)$ to indicate whether location i is selected ($z_i = 1$) or not ($z_i = 0$). An example is shown in Figure 1b. Building a warehouse at location i entails a finite and fixed setup cost f_i, and $\boldsymbol{f} = (f_1, f_2, \ldots, f_m)$. We temporarily assume warehouses have no capacity limitations, which will be relaxed in Section 6.2. We define another decision variable $\boldsymbol{y} = (y_1, y_2, \ldots, y_m)$ to represent the inventory levels at supply nodes. For simplicity of notations, we furthur assume there is only one kind of relief material, i.e., a relief package that contains food, clothes, water, etc. We will consider a multi-item case in Section 6.1 under mild assumptions. Each deployed product at supply node i entails a finite and fixed holding cost h_i, and we denote $\boldsymbol{h} = (h_1, h_2, \ldots, h_m)$. Additionally, $\Gamma(i)$ (and $\Gamma(j)$) is the set of adjacent nodes, which are directly connected to a specific node, for supply node i (and demand node j) when the network is $\mathcal{E}$ (see Figure 1c). Interchangeably, $\Gamma(i) = \{j : (i,j) \in \mathcal{E}\}$ and $\Gamma(j) = \{i : (i,j) \in \mathcal{E}\}$. We use



(a) Links: $\mathcal{E}$ (b) Links: $\mathcal{E}(\boldsymbol{z})$ (c) Adjacent Nodes: $\Gamma(i)$

Fig. 1: Notation Illustration

$\Gamma_{\boldsymbol{z}}(i)$ to emphasize the set of i's adjacent nodes when $\boldsymbol{z}$ is explicitly given. Warehouse $i \in \{i | z_i = 1, i \in \mathcal{W}\}$ can deliver products to all adjacent demand nodes $\Gamma_{\boldsymbol{z}}(i)$ via roads. Correspondingly, a demand node $j \in \mathcal{R}$ can receive products from its adjacent warehouses $\Gamma_{\boldsymbol{z}}(j)$ when the network structure is $\mathcal{E}(\boldsymbol{z})$. Suppose the decision maker has only limited information about demand, i.e., only the finite first moment $\boldsymbol{\mu}$, finite second moment $\boldsymbol{\Sigma}$, and support set $\mathbb{R}^n_+$ are known. Please note that, $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ can be estimated through tectonic analysis and geologic analysis by geologists. We also provide a data-driven approach based on estimating $\widehat{\boldsymbol{\mu}}$, $\widehat{\boldsymbol{\Sigma}}$ in Section 6.3.

In the second stage, after demand $\tilde{\boldsymbol{d}} = (\tilde{d}_1, \tilde{d}_2, \ldots, \tilde{d}_n)$ is realized, the decision maker can allocate predeployed items through network $\mathcal{E}(\boldsymbol{z})$ to satisfy as much of the demand as possible (or equivalently to minimize unmet demand). Each unit of unmet demand will trigger a penalty cost p. p is the same across all demand nodes since human lives are equally important regardless of where he or she resides. For simplicity, we let $p = 1$ in following analysis. $\boldsymbol{f}$ and $\boldsymbol{h}$ are normalized accordingly with respect to p. We ignore the transshipment cost in the second stage. This is reasonable in our case. First, the penalty cost

caused by unmet demand is far larger than the transshipment cost due to the importance of human lives. As long as there is a demand, transshipment should be conducted to fulfil the demand and save lives. Second, compared with the long-term prepositioning cost in the preparedness stage, the impact caused by the transshipment cost on the here-and-now decisions could be ignored. Therefore, we do not take transshipment into consideration. The ultimate objective is to minimize the total cost, including the setup cost, holding cost and the worst-case expected second-stage penalty cost.

Now, we are ready to model the disaster relief network design problem under demand distribution uncertainty. The formulation can be expressed as follows:

$$\begin{aligned}(DRND)\ \min_{\boldsymbol{y},\boldsymbol{z}}\ & \kappa(\boldsymbol{f}^T\boldsymbol{z}+\boldsymbol{h}^T\boldsymbol{y})+\sup_{\mathbb{P}\in\mathscr{F}(\boldsymbol{\mu},\boldsymbol{\Sigma})}\mathbb{E}_{\mathbb{P}}\left[g(\boldsymbol{y},\boldsymbol{z},\tilde{\boldsymbol{d}})\right]\\ \text{s.t. }\ & \boldsymbol{y}\le U\boldsymbol{z};\\ & \boldsymbol{y}\in\mathbb{R}_+^m,\boldsymbol{z}\in\{0,1\}^m,\end{aligned}$$

where the distribution ambiguity set $\mathscr{F}$, given finite mean $\boldsymbol{\mu}$, finite covariance matrix $\boldsymbol{\Sigma}$, and non-empty distribution family $\mathcal{P}(\mathbb{R}_+^n)$, is:

$$\mathscr{F}(\boldsymbol{\mu},\boldsymbol{\Sigma})=\left\{\mathbb{P}\in\mathcal{P}\left(\mathbb{R}_+^n\right)\left|\begin{array}{l}\tilde{\boldsymbol{d}}\sim\mathbb{P}\\ \mathbb{E}_{\mathbb{P}}[\tilde{\mathbf{d}}]=\boldsymbol{\mu}\\ \mathbb{E}_{\mathbb{P}}[\tilde{\boldsymbol{d}}\tilde{\boldsymbol{d}}^T]=\boldsymbol{\Sigma}\end{array}\right.\right\}.$$

In (DRND), $\boldsymbol{y}$ and $\boldsymbol{z}$ are the first-stage decision variables, representing the inventory levels and indicators of building warehouse decisions, respectively. $\kappa\ge 0$ is a risk attitude parameter that balances the costs of the two stages, and U is a sufficiently large positive number. The first constraint guarantees that relief packages can be stored only at locations where a warehouse is established; the remaining constraints are standard nonnegative and binary constraints. $g(\boldsymbol{y},\boldsymbol{z},\tilde{\boldsymbol{d}})$, the penalty cost of unmet demand, is obtained by solving a classic allocation problem after demand $\tilde{\boldsymbol{d}}$ is realized. Note that, without loss of generality, the unit penalty cost of each unit of unmet demand is normalized to one. Therefore, given $\boldsymbol{y}$, $\boldsymbol{z}$ and $\tilde{\boldsymbol{d}}$, the second-stage problem can be defined as:

$$g(\boldsymbol{y},\boldsymbol{z},\tilde{\boldsymbol{d}})=\min_{\boldsymbol{t}\in\Omega(\boldsymbol{y},\boldsymbol{z})}\sum_{j\in[n]}\tilde{d}_j-\sum_{(i,j)\in\mathcal{E}(\boldsymbol{z})}t_{ij},\tag{1}$$

where $\Omega(\boldsymbol{y},\boldsymbol{z})$ is the feasible region characterized by

$$\Omega(\boldsymbol{y},\boldsymbol{z}):=\{\boldsymbol{t}\in\mathbb{R}_+^{|\mathcal{E}(\boldsymbol{z})|}\big|\sum_{i:(i,j)\in\mathcal{E}(\boldsymbol{z})}t_{ij}\le\tilde{d}_j,\forall j\in[n];\sum_{j:(i,j)\in\mathcal{E}(\boldsymbol{z})}t_{ij}\le y_i,\forall i\in[m]\}.$$

$\mathcal{E}(\boldsymbol{z})$ represents the accessible roads based on decision $\boldsymbol{z}$, and t_{ij} is the amount of relief packages delivered from location i to demand node j. The first constraint of $\Omega(\boldsymbol{y},\boldsymbol{z})$ ensures that the total amount of items delivered to demand node j does not exceed the demand, and the second constraint is a capacity constraint for each warehouse i.

Solving Model (DRNR) can generate the desired network. However, before proceeding to the formal analysis, we remark that the optimization problem is challenging to solve in the following aspects.

- First, the first-stage decisions involve integer decision variable $\boldsymbol{z}$, whose change will change the dimension of feasible region $\Omega(\boldsymbol{y},\boldsymbol{z})$, making it impossible to directlys solve with commercial solvers.
- Second, evaluating the worst-case expected penalty cost is challenging due to the nested optimization structure.
- Third, the problem is generally a mixed-integer problem, solving which takes much computational time.

All these features make our network design problem challenging. In the next section, we first remove the impact $\boldsymbol{z}$ imposes on the dimension of $\Omega(\boldsymbol{y}, \boldsymbol{z})$. Then, we develop a conic approach using copositive programming to evaluate the worst-case expected penalty cost. Last, a branch-and-pricing method with warm start is proposed to accelerate the solving process.

4 Analysis and Reformulation

4.1 Preliminary Analysis

Clearly, Model (1) depends on both $\boldsymbol{y}$ and $\boldsymbol{z}$, and the accessible network varies under different $\boldsymbol{z}$, which prevents us from solving (DRND). Fortunately, the dependency on $\boldsymbol{z}$ is not necessary as shown in Proposition 1.

Proposition 1 *Given a feasible solution $(\boldsymbol{y}, \boldsymbol{z})$ to problem (DRND), for any demand realization $\tilde{\boldsymbol{d}} \in \mathbb{R}_+^n$, the value of* (1) *is equal to the objective value of the following problem.*

$$g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) = \min_{\boldsymbol{t} \in \Omega(\boldsymbol{y}, \tilde{\boldsymbol{d}})} \sum_{j \in [n]} \tilde{d}_j - \sum_{(i,j) \in \mathcal{E}} t_{ij}, \tag{2}$$

where

$$\Omega(\boldsymbol{y}, \tilde{\boldsymbol{d}}) := \{\boldsymbol{t} \in \mathbb{R}_+^r \big| \sum_{i:(i,j) \in \mathcal{E}} t_{ij} \leq \tilde{d}_j, \forall j \in [n]; \sum_{j:(i,j) \in \mathcal{E}} t_{ij} \leq y_i, \forall i \in [m]\}.$$

Proof See A.1

The intuition of the proof is that when there is a link but no warehouse is established on the supply node, one can always force the flows t_{ij} that originate from location i to zero. According to Proposition 1, instead of considering $(\boldsymbol{y}, \boldsymbol{z})$, we need to take only $\boldsymbol{y}$ into consideration. Removing $\boldsymbol{z}$ does not harm the optimality but comes at the cost of increasing the dimension of the recourse decision variables $\boldsymbol{t}$. Surprisingly, additional variables make further reformulation easier and provide greater insight into the selection of locations for warehouses, accelerating the solution process. The main intuition is that the shadow price of $\boldsymbol{t}$ in (2) evaluates importances of links of bipartite $\mathcal{E}$ rather than a subset $\mathcal{E}(\boldsymbol{z})$ in (1). Therefore, the importance of warehouse locations can be quantified. For example, it can be measured by aggregating the value of each link that originates from the location. We will further explain the details in Section 4.3.

Hereafter, we replace $g(\boldsymbol{y}, \boldsymbol{z}, \tilde{\boldsymbol{d}})$ with $g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$. However, since the feasible region $\Omega(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ depends on both $\boldsymbol{y}$ and $\tilde{\boldsymbol{d}}$, impeding the evaluation of $\mathbb{E}_{\mathbb{P}}[g(\boldsymbol{y}, \tilde{\boldsymbol{d}})]$, Model (DRND) is still intractable. We next employ max-flow min-cut theory to decouple the dependency as shown in Proposition 2.

Proposition 2 *Given $\boldsymbol{y}$ and $\tilde{\boldsymbol{d}}$, the second-stage allocation problem $g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ can be reformulated as follows,*

$$g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) = \max_{(\hat{\boldsymbol{u}}, \boldsymbol{v}) \in \hat{\mathcal{L}}_{lp} \cap \{0,1\}^{n+m}} \tilde{\boldsymbol{d}}^T \hat{\boldsymbol{u}} - \boldsymbol{y}^T \boldsymbol{v}, \tag{3}$$

where

$$\hat{\mathcal{L}}_{lp} := \left\{ (\hat{\boldsymbol{u}}, \boldsymbol{v}) \in \mathbb{R}_+^{n+m} \left| \begin{array}{l} v_i \geq \hat{u}_j, \forall (i,j) \in \mathcal{E} \\ \hat{\boldsymbol{u}} \leq \mathbf{1} \\ \boldsymbol{v} \leq \mathbf{1} \end{array} \right. \right\}.$$

Proof See A.2.

Proposition 2 generally states that, to obtain the value of $g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$, instead of searching for $\boldsymbol{t}$ in space $\Omega(\boldsymbol{y}, \tilde{\boldsymbol{d}})$, we can search for $(\hat{\boldsymbol{u}}, \boldsymbol{v})$ in space $\hat{\mathcal{L}}_{lp} \cap \{0,1\}^{n+m}$. Intuitively, Proposition 2 transfers the variables $\boldsymbol{y}$ and $\tilde{\boldsymbol{d}}$ from the right-hand side of inequalities in $\Omega(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ to the objective function of (3), leaving the new feasible region $\hat{\mathcal{L}}_{lp}$ irrelative to any variables. This reformulation is critical since $\hat{\mathcal{L}}_{lp}$ is deterministic, making the evaluation of $\mathbb{E}_{\mathbb{P}}[g(\boldsymbol{y}, \tilde{\boldsymbol{d}})]$ easier.

Now, we consider a reduced disaster relief network design (rDRND) model with a shortened $g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ in the form of (3) as follows:

$$\begin{aligned} (rDRND) \min_{\boldsymbol{y},\boldsymbol{z}} \ & \kappa(\boldsymbol{f}^T \boldsymbol{z} + \boldsymbol{h}^T \boldsymbol{y}) + \sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu}, \boldsymbol{\Sigma})} \mathbb{E}_{\mathbb{P}} \left[g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) \right] \\ \text{s.t.} \ & \boldsymbol{y} \leq U\boldsymbol{z}; \\ & \boldsymbol{y} \in \mathbb{R}^m_+, \boldsymbol{z} \in \{0,1\}^m. \end{aligned} \tag{4}$$

With a more tractable recourse problem, we next equivalently reformulate Model (4) as a mixed-integer conic programming problem. Apparently, the difficulty of analyzing model (rDRND) comes from evaluating the worst-case expected second-stage objective value. Therefore, we first dive into the second-stage problem. Denote the worst-case expected second-stage problem as

$$L(\boldsymbol{y}) := \sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu}, \boldsymbol{\Sigma})} \mathbb{E}_{\mathbb{P}} \left[g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) \right]. \tag{5}$$

Unfortunately, although the feasible region of $g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$, in the form of (3), is fixed and not related to the first-stage decision $\boldsymbol{y}$, the difficulty of evaluating (5) is not reduced. Indeed, Proposition 3 shows that problem (5) is at least NP-hard.

Proposition 3 *Given $g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ in the form of* (3)*, calculating the solution to problem* (5) *is at least NP-hard.*

Proof See A.3

While the problem is at least NP-hard, we can reformulate it as a conic programming problem based on copositive cone under a mild assumption, and then solve it via positive semidefinite relaxation.

4.2 A Conic Approach based on Copositive Cone

In this section, we reformulate problem (rDRND) as a mixed integer conic problem. Our reformulation heavily relies on conic programming, especially copositive cone programming. Note that we have provided a brief introduction to copositive cone programming in Section 2.2. For a deep understanding of copositive cones and a comprehensive review about copositive programming, we refer interested readers to Dickinson (2013) and Li et al. (2014). Now, we present one of the main results in this paper.

Proposition 4 *Given that the moment matrix $\begin{pmatrix} 1 & \boldsymbol{\mu}^T \\ \boldsymbol{\mu} & \boldsymbol{\Sigma} \end{pmatrix}$ lies in the interior of a $(1+n)\times(1+n)$-dimensional completely positive cone, problem (rDRND) can be equivalently reformulated as a mixed-integer conic*

problem (CO) based on copositive cones as

$$
\begin{aligned}
(CO)\ L_{CO} = \min_{\boldsymbol{y},\boldsymbol{z},\alpha_i,\beta_i,\theta_j,\tau,\boldsymbol{\xi},\boldsymbol{\Xi}} & \boldsymbol{f}^T\boldsymbol{y} + \boldsymbol{h}^T\boldsymbol{z} + \sum_{i\in[M]}(b_i\alpha_i + b_i^2\beta_i) + \tau + 2\boldsymbol{\mu}^T\boldsymbol{\xi} + \boldsymbol{\Sigma}\bullet\boldsymbol{\Xi} \\
s.t. \quad & \boldsymbol{y} \le U\boldsymbol{z}; \\
& \boldsymbol{y} \ge \boldsymbol{0}, \boldsymbol{z} \in \{0,1\}^m; \\
& \begin{pmatrix} \tau & \boldsymbol{\xi}^T & \frac{1}{2}\left(\sum_{i\in[M]}\boldsymbol{a}_i\alpha_i - \boldsymbol{\theta}\right)^T \\ \boldsymbol{\xi} & \boldsymbol{\Xi} & \boldsymbol{0} \\ \frac{1}{2}\left(\sum_{i\in[M]}\boldsymbol{a}_i\alpha_i - \boldsymbol{\theta}\right) & \boldsymbol{0} & \sum_{i\in[M]}\boldsymbol{a}_i\boldsymbol{a}_i^T\beta_i + \boldsymbol{Diag}(\boldsymbol{\theta}) \end{pmatrix} \\
& +\frac{1}{2}\begin{pmatrix} \boldsymbol{0} & \boldsymbol{0} & \begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{y}\\ \boldsymbol{0}\end{bmatrix}^T \\ \boldsymbol{0} & \boldsymbol{0} & \begin{bmatrix}-\boldsymbol{E}^{[n]}\\ \boldsymbol{0}\end{bmatrix}^T \\ \begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{y}\\ \boldsymbol{0}\end{bmatrix} & \begin{bmatrix}-\boldsymbol{E}^{[n]}\\ \boldsymbol{0}\end{bmatrix} & \boldsymbol{0} \end{pmatrix} \succeq_{co} \boldsymbol{0},
\end{aligned}
$$

Proof See A.4

In Model (CO), $\boldsymbol{y} \in \mathbb{R}_+^m$, $\boldsymbol{z} \in \{0,1\}^m$, $\boldsymbol{\alpha} \in \mathbb{R}^M$, $\boldsymbol{\beta} \in \mathbb{R}^M$, $\boldsymbol{\theta} \in \mathbb{R}^N$, $\tau \in \mathbb{R}$, $\boldsymbol{\xi} \in \mathbb{R}^n$, and $\boldsymbol{\Xi} \in \mathbb{R}^{n\times n}$ are decision variables. $N = 2m + 2n + r$ and $M = m + n + r$. $\boldsymbol{a}_i, \forall i \in [M]$ is the coefficient vector of the i-th equation in equation system $\hat{\mathcal{L}}_{lp}^{\dagger}$ equivalently derived from $\hat{\mathcal{L}}_{lp}$:

$$
\hat{\mathcal{L}}_{lp}^{\dagger} := \left\{ (\hat{\boldsymbol{u}}, \boldsymbol{v}, \boldsymbol{s}^{\dagger}, \hat{\boldsymbol{u}}^{\dagger}, \boldsymbol{v}^{\dagger}) \in \mathbb{R}_+^N \,\middle|\, \begin{array}{l} \hat{u}_j - v_i + s_{ij}^{\dagger} = 0, \forall (i,j) \in \mathcal{E} \\ \hat{\boldsymbol{u}} + \hat{\boldsymbol{u}}^{\dagger} = \boldsymbol{1} \\ \boldsymbol{v} + \boldsymbol{v}^{\dagger} = \boldsymbol{1} \end{array} \right\}.
$$

Interestingly, the first r constraints of $\hat{\mathcal{L}}_{lp}^{\dagger}$ involve only roads in $\mathcal{E}$, and the following n and m constraints involve only demand nodes and supply nodes, respectively. $\boldsymbol{E}^{[n]}$ is an n-dimensional identity matrix. $\boldsymbol{Diag}(\boldsymbol{\theta})$ represents a diagonal matrix with the i-th diagonal element equal to θ_i. The first two constraints are the standard big-M constraint and nonnegative constraint, where U is a sufficiently large positive number. The third constraint requires that the resulting matrix after the addition operation lies in a copositive cone with proper dimensions. Please note that although we reformulate the original problem into a conic problem, the computational complexity only shifts from finding the worst-case distribution to finding a satisfied copositive cone and is therefore not reduced.

4.3 Acceleration of Branch-and-Pricing

Although we have reformulated (rDRND) as a convex conic problem based on a copositive cone, Model (CO) still cannot be efficiently solved because of the complexity of the copositive cone and the binary decision variable $\boldsymbol{z}$. The former issue can be addressed via semidefinite positive relaxation, while the latter issue is not trivial, and brute-force search is not suitable due to exponentially many solutions. In

this subsection, we adopt the classic branch-and-pricing framework with depth-first search to search for optimal solutions. To accelerate the branch-and-pricing procedure, we develop a warm-start approach that exploits known information over moments and the results from Scarf (1958) to obtain an interpretable high-quality initial solution. Additionally, we propose a heuristic to quantify the value of each location in each branch-and-pricing iteration, thus accelerating the solving process by branching on a better variable. As the only integer decision variable is $\boldsymbol{z}$, the proposed branch-and-pricing technique is only applied only to $\boldsymbol{z}$.

Lemma 1 *(Scarf (1958)) Suppose only demand mean μ and variance σ are known in advance; then a distributionally robust newsvendor problem to maximize the expected revenue can be formulated as follows,*

$$\max_{q} -hq + \inf_{\theta \sim \mathcal{F}(\mu,\sigma)} \mathbb{E}_{\theta}[p \min\{\tilde{d}, q\}]$$

where h and p are the unit holding cost and selling price for each item. The optimal quantity q^* is

$$q^* = \begin{cases} \mu + \frac{\sigma}{2}\left(\sqrt{\frac{p-h}{h}} - \sqrt{\frac{h}{p-h}}\right), & if\ \frac{\mu}{\sigma} \geq \frac{h}{p-h}; \\ 0, & otherwise. \end{cases} \tag{6}$$

In our problem, we can regard each demand node $j \in \mathcal{R}$ as an independent newsvendor facing the same problem. If the relief material of demand node j is supplied by warehouse location i, then the holding cost is h_i, and the unit revenue by fulfilling demand is $p_j = 1, \forall j \in \mathcal{R}$. Therefore, if newsvendor j stores the inventory at location i, the optimal inventory level is q^*_{ij}:

$$q^*_{ij} = \begin{cases} \mu + \frac{\sigma}{2}\left(\sqrt{\frac{p_j-h_i}{h_i}} - \sqrt{\frac{h_i}{p_j-h_i}}\right), & if\ \frac{\mu}{\sigma} \geq \frac{h_i}{p_j-h_i}; \\ 0, & otherwise, \end{cases} \tag{7}$$

where $p_j = 1, \forall j \in \mathcal{R}$.

Proposition 5 (Warm-start policy) *Given the optimal inventory level q^*_{ij}, we can obtain an initial solution $\boldsymbol{z}^{init}$ by solving the following assignment problem,*

$$\begin{aligned} \min_{\boldsymbol{z},\boldsymbol{t}} \quad & \boldsymbol{f}^T\boldsymbol{z} + \sum_{(i,j)\in\mathcal{E}} h_i q^*_{ij} t_{ij} \\ s.t. \quad & \sum_{i\in[n]} t_{ij} \geq 1,\ \forall j \in [j]; \\ & t_{ij} \leq z_i,\ \forall (i,j) \in \mathcal{E}; \\ & \boldsymbol{z} \in \{0,1\}^m, \boldsymbol{t} \in \{0,1\}^r. \end{aligned} \tag{8}$$

Furthermore, $\boldsymbol{z}^{init}$ can provide an upper bound for problem (CO), i.e., $L_{CO}(\boldsymbol{z}^{init}) \geq L_{CO}$, where $L_{CO}(\boldsymbol{z}^{init})$ is objective value of problem (CO) by fixing the variable $\boldsymbol{z} = \boldsymbol{z}^{init}$.

In this case, we manually ignore the correlated demand uncertainty between demand nodes. The decision variables are building warehouses decisions $\boldsymbol{z}$ and the assignment decisions $\boldsymbol{t}$. The objective is the cost of building warehouses plus the total holding cost. The first constraint requires that each demand node should be assigned to at least one warehouse. The second constraint requires that demand node j could be assigned to warehouse i only when warehouse i is established. The remaining constraints are binary requirements. Solving the problem leads to a warm-start decision $\boldsymbol{z}^{init}$, since $\boldsymbol{z}^{init}$ represents the warehouse decisions when correlated demand uncertainty is manually ignored. Although the quality of

the warm-start solution $\boldsymbol{z}^{init}$ is not theoretically guaranteed, we will numerically demonstrate that this solution can help reduce the computational time significantly.

In our branch-and-pricing framework, in addition to determining a high-quality initial solution, another approach to accelerating the solving process is selecting a good variable to branch by appropriately pricing the value of each location variable. We apply an interpretable heuristic to produce the next branch. Note that the decision variable $\alpha_k, \forall k \in [r]$ in Model (CO) is the dual variable to the first-r road constraints of $\hat{\mathcal{L}}_{lp}^{\dagger}$. According to duality theory, economically, the solution $\alpha_k^*, \forall k \in [r]$ can be interpreted as the value of the corresponding road in the current iteration. Therefore, at each branch-and-pricing node, we can approximately evaluate the value of each location as the aggregated value of links that originate from it, and then select the most valuable location to branch. To describe our idea in a clearer mathematical manner, we first define an index mapping from road (i, j) to a scalar index, representing the road index in $[r]$. For example, we use subscript $_{(i,j)}, \forall (i, j) \in \mathcal{E}$ to represent road (i, j)'s index in set $[r]$. Now, we are ready to formally present our pricing policy in the Proposition 6.

Proposition 6 (Pricing policy) *During each iteration, after obtaining $\alpha_{(i,j)}^*, \forall (i, j) \in \mathcal{E}$ by solving Model (CO), the next supply location to branch is*

$$i^* = \arg \max_{i \in [m] \setminus \mathcal{C}} \alpha_{(i,\bullet)}. \tag{9}$$

where the value of location i is $\alpha_{(i,\bullet)} = \sum_{j \in \Gamma(i)} \alpha_{(i,j)}^*$, and $\mathcal{C}$ is the set of warehouse locations that have been selected to branch before this iteration.

Note that $\Gamma(i)$ is the set of arcs from the supply node i to the demand nodes. Therefore, $\alpha_{(i,\bullet)}$ can be seen as the value of the supply location i. Intuitively, we greedily select one supply location that has not been visited yet, having the highest aggregated value of all arcs that originate from it. Similar arguments have also been adopted in Yan et al. (2018). Formally, we summarize the proposed accelerated branch-and-pricing algorithm as Algorithm 1. We conduct further numerical experiments to illustrate the effect on reducing the computational time.

Algorithm 1 Accelerated Branch-and-Pricing Algorithm

```
Set z^init by solving (8)                                   ▷ Proposition 5
Solve (CO) with z^init to get L^init
Initialize z* ← z^init, L* ← L^init
Initialize constraint set C on z, C := {0 ≤ z ≤ 1}
Run DFS(C)

function DFS(C)
    Solve linear relaxation (CO) with C to obtain current objective value L
    if L < L* then
        if z is integer then
            L* ← L, z* ← z
        else
            Select supply location i according to (9)           ▷ Proposition 6
            C_1 ← C ∪ {z_i = 1}
            Run DFS(C_1)
            C_0 ← C ∪ {z_i = 0}
            Run DFS(C_0)
        end if
    end if
end function
```

5 Numerical Study

In this section, we conduct numerous experiments to examine the advantages of explicitly considering the first two moments of demand information in disaster relief network design problems. First, we run our models on synthetic datasets to scrutinize the average performance and the model's robustness to initial network structures. Sensitivity analyses are conducted to examine the influences of parameters. Second, a case study of Ya'an Earthquake that happened in 2013 is examined to demonstrate the superiority of our proposed model in practice. A stochastic model and a mean-variance model, which will be discussed in detail in the next subsection, are introduced as benchmarks throughout this section. We use positive semidefinite relaxation to solve copositive programming problems. Specifically, we use $\{\boldsymbol{A} \,|\, \boldsymbol{A} = \boldsymbol{A}_1 + \boldsymbol{A}_2, \boldsymbol{A}_1 \succeq \boldsymbol{0}, \boldsymbol{A}_2 \geq \boldsymbol{0}\}$ as an inner approximation to the copositive cone. Experiments are conducted on a computer with 16 GB memory and an AMD 3700X CPU. We use Mosek 9.2 to solve conic problems and Gurobi 9.0.3 to solve mixed-integer problems. All codes are open-source[1], and interested readers can easily reproduce the results of our experiments.

5.1 benchmark

We introduce a two-stage stochastic model and a mean-variance distributionally robust model as benchmarks. The former model fully exploits all demand samples, while the latter considers only two statistics, mean and variance, and does not consider correlations between demand nodes. Since our model exploits only the first two moments, the amount of available information is slightly more than that of the mean-variance model but less than that of the stochastic model.

SAA with Empirical Distribution

When demand is random with an unknown distribution $\mathbb{P}'$ supported by $\mathbb{R}^n_+$, and a set of historical observed demand containing N samples is available, the network design problem can be formulated as a two-stage stochastic model as follows:

$$
\begin{aligned}
(SAA) \min_{\boldsymbol{y},\boldsymbol{z}} \; & \kappa(\boldsymbol{f}^T\boldsymbol{z} + \boldsymbol{h}^T\boldsymbol{y}) + \frac{1}{N}\sum_{i=1}^{N} g(\boldsymbol{y}, \tilde{\boldsymbol{d}}_i) \\
\text{s.t.} \; & \boldsymbol{y} \leq U\boldsymbol{z}; \\
& \boldsymbol{y} \in \mathbb{R}^m_+, \boldsymbol{z} \in \{0,1\}^m,
\end{aligned}
$$

where $g(\boldsymbol{y}, \tilde{\boldsymbol{d}}_i)$ is the objective value of recourse allocation problems, which is the same as (2) but with the i-th historical demand sample $\tilde{\boldsymbol{d}}_i$. The first term of the objective function is the total setup cost, including the warehouse setup cost $\boldsymbol{f}^T\boldsymbol{z}$ and the holding cost $\boldsymbol{h}^T\boldsymbol{y}$. The second term represents the average cost on historical demand samples when the inventory level is $\boldsymbol{y}$, which converges to $\mathbb{E}_{\mathbb{P}'}[g(\boldsymbol{y}, \tilde{\boldsymbol{d}})]$ as N goes to infinity according to the Law of Large Number. $\boldsymbol{y}$ and $\boldsymbol{z}$ are the inventory and location selection decision variables, respectively. The feature that distinguishes Model (SAA) from Model (rDRND) is how we evaluate the expected second-stage objective value.

Mean-Variance

The other benchmark is a distributionally robust model whose ambiguity set depends only on finite mean $\boldsymbol{\mu}$ and finite variance $\boldsymbol{\sigma}^2$. In other words, the covariance is ignored. In such a case, the decision maker attempts to evaluate the expected second-stage cost under the worst-case distribution. Formally,

[1] Source code is available at GitHub: Hyperlink is temporarily disabled for peer review

the model is:

$$
\begin{aligned}
(MV)\ \min_{\boldsymbol{y},\boldsymbol{z}}\ & \kappa(\boldsymbol{f}^T\boldsymbol{z}+\boldsymbol{h}^T\boldsymbol{y}) + \sup_{\mathbb{P}\in\mathscr{F}_{mv}(\boldsymbol{\mu},\boldsymbol{\sigma}^2)} \mathbb{E}_{\mathbb{P}}\left[g(\boldsymbol{y},\tilde{\boldsymbol{d}})\right] \\
\text{s.t. } & \boldsymbol{y}\le U\boldsymbol{z}; \\
& \boldsymbol{y}\in\mathbb{R}^m_+, \boldsymbol{z}\in\{0,1\}^m,
\end{aligned}
$$

where the ambiguity set $\mathscr{F}_{mv}(\boldsymbol{\mu},\boldsymbol{\sigma}^2)$ is

$$
\mathscr{F}_{mv}(\boldsymbol{\mu},\boldsymbol{\sigma}^2) = \left\{ \mathbb{P}\in\mathcal{P}\left(\mathbb{R}^n_+\right) \left| \begin{array}{l} \tilde{\boldsymbol{d}}\sim\mathbb{P} \\ \mathbb{E}_{\mathbb{P}}[\tilde{\boldsymbol{d}}]=\boldsymbol{\mu} \\ \mathbb{E}_{\mathbb{P}}[(\tilde{d}_j-\mu_j)^2]=\sigma_j^2,\ \forall j\in\mathcal{R} \end{array} \right. \right\},
$$

which is assumed to be non-empty. The objective function of Model (MV) is almost the same as that of Model (rDRND) except for the slightly different ambiguity set. Although Model (MV) is still intractable in most cases, we can transform $\mathscr{F}_{mv}(\boldsymbol{\mu},\boldsymbol{\sigma}^2)$ into $\mathscr{F}(\boldsymbol{\mu},\boldsymbol{\Sigma}_{mv})$, where $\boldsymbol{\Sigma}_{mv}=\boldsymbol{\mu}\boldsymbol{\mu}^T+\boldsymbol{Diag}(\boldsymbol{\sigma}^2)$. Then, every analysis holds, as shown in Proposition 4. Replacing $\boldsymbol{\Sigma}$ in Model (CO) with $\boldsymbol{\Sigma}_{mv}$ yields the resulting conic model, which is solvable via relaxations. One noteworthy feature of Model (MV) is that as $\mathscr{F}(\boldsymbol{\mu},\boldsymbol{\Sigma})\subseteq\mathscr{F}(\boldsymbol{\mu},\boldsymbol{\Sigma}_{mv})$, it would be more conservative than Model (CO) is, resulting in a higher expected second-stage cost under the same $\boldsymbol{y}$ and $\boldsymbol{z}$. Nevertheless, since both Model (CO) and Model (MV) balance setup costs and second-stage penalty costs, it is hard to compare the amount of inventory levels and, further, the total cost at this point.

5.2 Experiments on Synthetic Datasets

5.2.1 Data Generation

Consider a network with m potential warehouses and n demand nodes. We randomly generate dozens or hundreds of networks, and each network is defined by a six-element tuple $(m,n,\mathcal{E},\boldsymbol{f},\boldsymbol{h},\boldsymbol{\mu})$. m and n are the numbers of potential warehouse nodes and demand nodes, and $\mathcal{E}$ is a set of links. According to Ni et al. (2018), the average node degree of a typical real-world road network is approximately 2.4; to mimic reality, we therefore require $|\mathcal{E}|=1.2(m+n)$. The following two constraints on the generated network must satisfied: (1) at least one link is connected to each supply node and (2) at least one link is connected to each demand node. These conditions help to avoid the appearance of bipartite graphs with isolated nodes. $\boldsymbol{f}$ is the fixed setup cost, which is randomly drawn from a uniform distribution U[10, 25], $\boldsymbol{h}$ is the holding cost drawn from U[0.1, 0.2], and $\boldsymbol{\mu}$ is the first moment, i.e., the mean, of demand, drawn from U[400, 600].

Initially, we set the correlation coefficient between each demand node pair as $\rho=0.3$, coefficient of variation of each demand node as $c_v=0.3$, and risk attitude coefficient $\kappa=1.0$. Therefore, the standard deviation of each demand node is $\sigma_j=\mu_j c_v, \forall j\in[n]$. For simplicity, we assume the correlation parameter ρ and coefficient of variation c_v are applied to all demand node pairs or demand nodes, which means the covariance of any two demand nodes (i,j) is $\sigma_i\sigma_j\rho,\ \forall i\in[m], j\in[n]$. Therefore, the covariance matrix of $\tilde{\boldsymbol{d}}$ is $\boldsymbol{\Sigma}_{\rho,c_v}$ whose (i,j)-position element is $\sigma_i\sigma_j\rho$. We use the subscripts $_{\rho,c_v}$ to emphasize that the covariance matrix depends on ρ and c_v. We randomly draw 50 demand samples from a truncated multivariate Gaussian distribution $N(\boldsymbol{\mu},\boldsymbol{\Sigma}_{\rho,c_v})_+$ with domain $\mathbb{R}^n_+$, mean $\boldsymbol{\mu}$, and covariance matrix $\boldsymbol{\Sigma}_{\rho,c_v}$. Based on 50 samples, we calculate the sampled mean $\hat{\boldsymbol{\mu}}$, sampled variance $\hat{\boldsymbol{\sigma}}_j$ and sampled covariance matrix $\hat{\boldsymbol{\Sigma}}_{\rho,c_v}$. These sampled statistics are the input parameters of the compared models.

After obtaining optimal solutions by solving the three models, we conduct the second-stage simulations on two independently generated datasets. The first dataset of 1000 demand realizations is generated from

$N(\boldsymbol{\mu}, \boldsymbol{\Sigma}_{\rho,c_v})_+$, and the second dataset of another 1000 demand realizations is from a mixture distribution consisting of four sub-distributions, including a two-point distribution $\mathbb{P}(\tilde{\boldsymbol{d}} = \boldsymbol{\mu}-\boldsymbol{\sigma}) = \mathbb{P}(\tilde{\boldsymbol{d}} = \boldsymbol{\mu}+\boldsymbol{\sigma}) = 0.5$, element-wise independent uniform distribution $U\left[\boldsymbol{\mu} - \sqrt{3}\boldsymbol{\sigma}, \boldsymbol{\mu} + \sqrt{3}\boldsymbol{\sigma}\right]$, independent normal distribution $d_j \sim N(\mu_j, \sigma_j), \forall j \in [n]$, and log-normal distribution with mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$. The above four distributions are truncated to $\mathbb{R}^n_+$. It is easy to check that the mixture distribution ignores correlations but has a mean and standard deviation near the true $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$. We separately analyze the results for the truncated Multivariate Gaussian Distribution (MGD) and Mixture Distribution (MixD). For each network $(m, n, \mathcal{E}, \boldsymbol{f}, \boldsymbol{h}, \boldsymbol{\mu})$, we regard the average (or worst-case) performance of the two datasets as indicative of the corresponding network's performance.

5.2.2 Simulation Results

Following the initial setting in the above subsection, we first restrict ourselves to the $(m, n) = (8, 8)$ structure for a closer look at the model's performance. We randomly generate 200 networks with $(\rho, c_v, \kappa) = (0.3, 0.3, 1.0)$. Unless stated otherwise, all analyses in this subsection rely on the 200 networks: for each network, we construct a metrics based on the simulation results, for example, the average unmet demand, to represent the network's performance. Then, we calculate the average for the 200 networks.

First-stage Deployment

Table 1 summarizes the first-stage decisions and resulting networks. The first column, *total inv*, rep-

Table 1: Summary of the Designed Networks

	total inv.	# of w.h.	# of roads	w.h. degree	d.n. degree
SAA	4903.43	4.06	11.57	2.91	1.45
	(300.33)	(0.78)	(1.57)	(0.45)	(0.20)
CO	4944.17	3.79	10.96	2.96	1.37
	(201.06)	(0.72)	(1.52)	(0.48)	(0.19)
MV	4638.28	3.77	11.15	3.02	1.39
	(194.14)	(0.72)	(1.44)	(0.48)	(0.18)

resents the average total inventory of the 200 networks. The numbers in parentheses show the standard deviations. Clearly, Model (CO) results in the highest inventory level, while Model MV has the lowest inventory level. The difference results from whether the strong correlation of demand is taken into account. The second column, *# of w.h.*, reveals the number of established warehouses: on average, Model (SAA) selects almost half of the potential locations to build warehouses. The third column is the number of links in the designed networks: Model (CO) designs the sparsest network. The last two columns exhibit the warehouse node and demand node degrees in the designed network $\mathcal{E}(\boldsymbol{z})$.

Cost

We next investigate the cost performance, including the first-stage deployment cost and second-stage penalty due to unmet demand, which are the costs of greatest concern. Table 2 shows the average costs for the 200 networks under different models with MGD and MixD realizations. The numbers in parentheses represent the standard deviation. The first two columns represent the setup cost $\boldsymbol{f}$ and the holding cost $\boldsymbol{h}$. $\boldsymbol{p}$ is the penalty cost, or equivalently, the unsatisfied demand. *total cost* is the aggregated cost. Clearly, Model (CO) always achieves the lowest total cost. One remarkable decrease occurs for $\boldsymbol{p}$, which decreases from approximately 64.5 (for SAA) to 53.1 (for (CO)), i.e., a 17.7% decrease in unmet demand under the in-sample simulation. For the MixD simulation, the decrease is also considerable, from 70.0 (for SAA) to 57.3, approximately 18.4%, or from 108.7 (for MV) to 57.3, approximately 47.3%. Although the absolute decrease is marginal, the relative decrease is remarkable, which highlights the advantages of our proposed

Table 2: Cost Comparison, $(m, n) = (8, 8), (\rho, c_v, \kappa) = (0.3, 0.3, 1.0)$

			MGD		MixD	
	f	**h**	**p**	total cost	**p**	total cost
SAA	67.7	660.2	64.5	792.5	70.0	798.0
	(14.9)	(70.5)	(31.7)	(78.2)	(38.8)	(81.0)
CO	63.4	668.5	53.1	785.0	57.3	789.2
	(13.9)	(66.5)	(12.2)	(78.5)	(15.0)	(82.0)
MV	63.3	636.1	96.1	795.5	108.7	808.1
	(13.7)	(66.1)	(15.4)	(80.1)	(15.8)	(81.3)

conic model: we can further substantially reduce the unmet demand even though it is already small. This managerial insight is quite attractive, especially to disaster management experts or fulfilment-oriented warehouse managers, since their priority is to reduce unmet demand without increasing expenditure.

We next scrutinize the unmet demand more deeply. Figure 2 shows the distribution of the average unmet demand estimated by kernel density estimation (KDE), where the horizontal axis shows the average unmet demand obtained through simulations, and the vertical axis represents the probability.

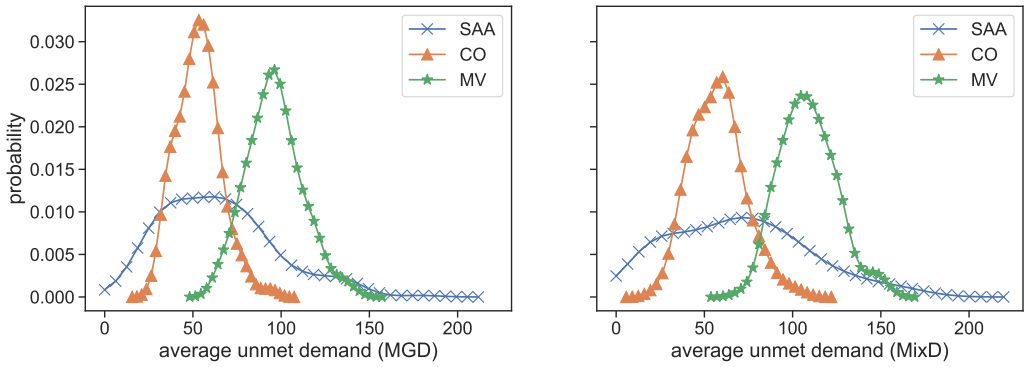


Fig. 2: Average unmet demand distribution, $(\rho, c_v, \kappa) = (0.3, 0.3, 1.0)$

In the left graph, when the true demand distribution aligns with the distribution that decision makers know, Model (CO) dominates Model (MV), except in some instances of overlap at approximately 75. Although Model (CO) achieves a higher level of concentration at approximately 55, the lowest unmet demand is achieved by Model (SAA), whose performance is more scattered, entailing several worst cases at 150 levels. The right graph shows the results from the MixD simulations. Similarly, Model (CO) dominates Model (MV) most of the time. However, the most interesting change is the increased dispersion of Model (SAA). Clearly, the performance of Model (SAA) becomes unstable in different networks, while (CO) and (MV) remain concentrated near their former mean values. The changes in the standard deviations of $\boldsymbol{p}$ in Table 2 further support our findings. Furthermore, the slimmer shapes of Model (CO) in both graphs demonstrate the robustness of our proposed model to initial network graphs in terms of unmet demand, since the average unmet demand of 200 graphs all centralizes at approximately 50.

Service Level

Another critical metric is the service level. Table 3 summarizes two kinds of service levels. The type 1 service level, or α service level, is an event-oriented performance criterion that measures the probability that the entire demand will be completely satisfied by stock on hand. The type 2 service level, or β service level, is a quantity-oriented performance measure describing the proportion of satisfied demand to total demand. Model (CO) performs slightly better than Model (SAA) in terms of the type 2 service

Table 3: Service Level, $(m, n) = (8, 8), (\rho, c_v, \kappa) = (0.3, 0.3, 1.0)$

	MGD		MixD	
service level (%)	type1	type2	type1	type2
SAA	94.79	98.76	93.75	98.59
CO	95.61	98.99	94.32	98.85
MV	93.82	98.16	93.10	97.85

level. Moreover, in terms of the type 1 service level, Model (CO) outperforms the two other models by approximately 1%.

Value-at-Risk

Furthermore, we calculate the 99.9th, 95th, 90th, 85th, and 80th quantiles of unmet demand to demonstrate the ability of the models to hedge long-tail risk. When the true distribution is known to decision makers, these quantiles can be regarded as estimators of the value-at-risk (VaR) of unmet demand after implementing the corresponding first-stage solutions. Table 4 shows the VaR values under different quantiles. *MGD VaR* shows that Model (CO) dominates other models under all circumstances, which indicates that its ability to hedge long-tail risk comes from information about the true underlying distribution. Surprisingly, columns *MixD VaR* in Table 4 further reveal that Model (CO) is sufficiently robust to the

Table 4: Unmet Demand with Various Quantiles

	MGD VaR					MixD VaR				
unmet demand	99.9%	95%	90%	85%	80%	99.9%	95%	90%	85%	80%
SAA	1323.3	406.4	212.8	111.0	56.3	798.3	335.4	325.9	250.0	119.8
CO	1280.0	350.9	163.2	71.8	27.5	761.8	267.9	265.0	224.8	97.5
MV	1585.0	617.8	365.5	206.2	100.2	968.8	569.7	569.7	398.3	171.5

misspecification in the distribution. In terms of the 95th quantile, Model (CO)'s unmet demand under MixD is 20% (from 335.4 to 267.9) lower than that of Model (SAA), while the gap is only 13.6% (from 406.4 to 350.9) under the MGD situation. This result verifies the advantages of Model (CO) and the distributionally robust technique, especially when historical data are limited and misspecification in the distribution is likely.

CPU Time

Computational time is also essential in practice. Although the problem we consider is a strategy-level problem, reasonable computational time is necessary. Therefore, we record the CPU time for solving each optimization problem, thereby highlighting the superiority of the proposed branch-and-pricing acceleration algorithm. Notably, since we run experiments with four parallel tasks synchronously on one computer, the CPU time is slightly longer. Table 5 summarizes the time needed to solve each model. We use *(acce)* to indicate that the model is solved with the accelerated branch-and-bound technique proposed in Section 4.3. The first column, *CPU Time (s)*, reports the computational time in seconds to obtain the optimal solution, and the second column, *Node*, exhibits the number of visited branch-and-bound nodes before

Table 5: CPU Time to Solve Models

	CPU Time (s)	Node
SAA	0.05	-
CO	748.79	98.80
CO (acce)	337.30	44.09
MV (acce)	355.55	43.88

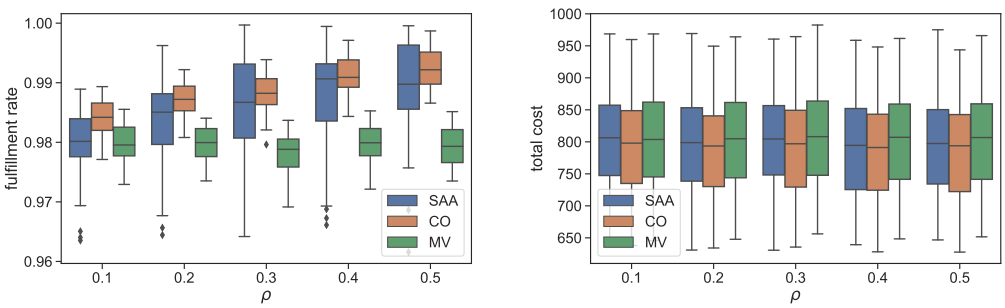
terminating the algorithm. Clearly, for Model (CO), the accelerated algorithm reduces the time by more than half by visiting fewer nodes. The decreased computational time also demonstrates that including more variables in the reformulated model in Proposition 1 is worthwhile.

5.2.3 Sensitivity Analysis

We modify some parameters (correlation parameter ρ, coefficient of variation c_v, and risk attitude κ) to conduct sensitivity analyses. Since we have explored the metrics of interest in detail, throughout this part, we consider only 50 graphs for each parameter combination. Additionally, we focus on MixD simulations, where the true underlying distribution deviates from the distribution used in the training sample generation process. This situation is likely to occur in reality; hence, a sufficiently robust relief material network is needed. The results show that all the conclusions proposed earlier hold. Moreover, paired t-tests on performances between Model (CO) and other two benchmarks reject the null hypothesis with p-value$\leq$0.001, implying that the improvement of Model (CO) is statistically significant.

Correlation Parameter ρ

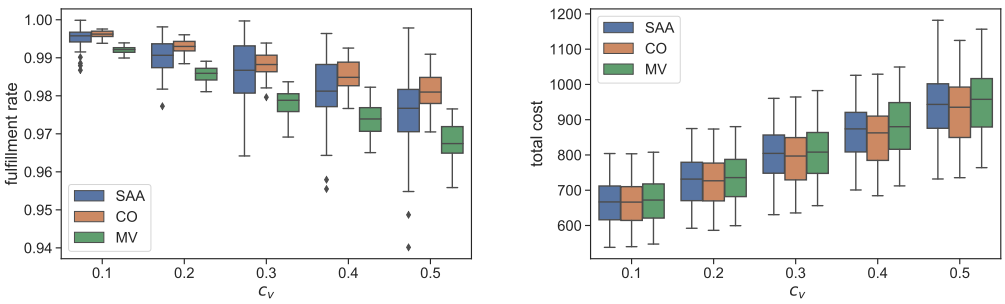
We first present a sensitivity analysis of the correlation parameter ρ, which we change from 0.1 to 0.5. Figure 3a shows the fulfilment rates of the three models under different values of parameter ρ. The



(a) Fulfilment Rate (b) Total Cost

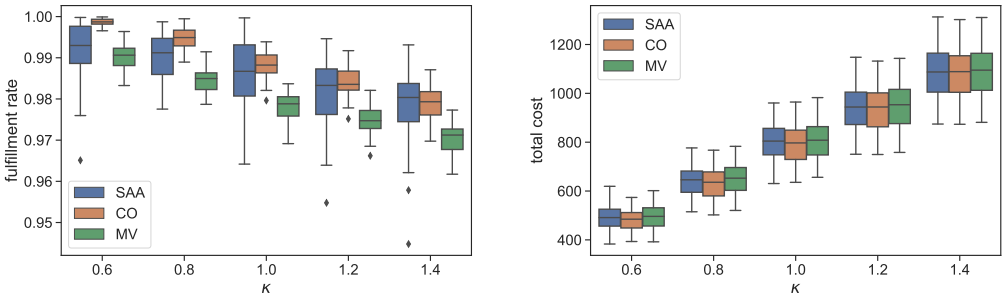
Fig. 3: Sensitivity Analysis of ρ

fulfilment rate, or type 2 service level, measures the proportion of demand that is satisfied. Model (CO) always has a higher fulfilment rate, on average, than the other models. Moreover, the perturbation of Model (CO) is much less severe than that of the other models, indicating that Model (CO) is robust to the network structure. An interesting trend is that the fulfillment rate of Model (MV) is stable over a range of ρ values, while that of the other models increases. The difference is due to the models' abilities



(a) Fulfillment Rate (b) Total Cost

Fig. 4: Sensitivity Analysis on c_v



(a) Fulfillment Rate (b) Total Cost

Fig. 5: Sensitivity Analysis of κ

to utilize correlation information. While Model (MV) does not consider any correlation between demand nodes, Model (CO) and Model (SAA) take the correlation into consideration explicitly and implicitly, respectively. Another noteworthy phenomenon is that outliers appear only in Model SAA, reflecting its weakness in terms of hedging extreme risks. Figure 3b exhibits the total cost of the three models under different values of ρ. Model (CO) can always achieve the lowest total cost.

Coefficient of Variation c_v

As in the analysis of ρ, we first consider the fulfilment rate. Figure 4a depicts fulfilment rates under a range of c_v values. As c_v increases, the mean fulfilment rates decrease in general, and the variance of the fulfillment rate increases. However, among the three models, Model (CO) has the lowest rate of decrease and the smallest variance, which further emphasizes the robustness of Model (CO). Figure 4b shows the change in total cost in terms of c_v. The total cost increases as c_v increases since more warehouses and products are needed to counter the increasing uncertainty.

Risk Attitude Parameter κ

Figure 5a represents the relationship between the fulfilment rate and risk attitude κ. As κ increases, i.e., the decision maker focuses more on the first-stage cost, the fulfilment rate decreases significantly.

This is a direct result of a change in risk attitude since a larger κ reflects a more aggressive attitude to future risk. Therefore, the corresponding prepositioning decisions become less conservative. Another noteworthy phenomenon is the diminishing performance gap between Model (CO) and Model (SAA) as κ increases. Since more attention is shifted to the deployment cost, the advantage of Model (CO) in addressing demand uncertainty is gradually weakened. Therefore, Model (CO)'s superiority in terms of fulfilment rate diminishes. Figure 5b expresses the total costs for the models as κ increases. Model (CO) has a slightly lower total cost, regardless of the value of κ.

To summarize this subsection, we conduct numerous experiments on given networks to evaluate the performance of the proposed copositive cone-based conic model. The conic model is demonstrated to be robust to demand uncertainty in most cases and outperforms the SAA-based model. Additionally, it is essential to incorporate correlation information into modelling for situations such as disaster management where demand correlation indeed exists and plays an important role. As the variance or correlation increases, these values become more influential in the subsequent second-stage fulfilment performance. Furthermore, we demonstrated that the acceleration algorithm indeed reduces the computation time. We also conduct simulations on other networks beyond (8, 8) structure, and leave the results and analysis to B.

5.3 Case Study: Ya'an Earthquake

Earthquakes are one of the major natural disasters in China. After being heavily impacted by the 2008 Wenchuan Earthquake, the same area was affected by the Ya'an Earthquake, a M_s 7.0 earthquake occurred at Ya'an County in Sichuan Province, PR China, in 2013. The earthquake caused injuries of approximately 12,000 people injured and massive economic losses. The affected areas are depicted in Figure 6. In our

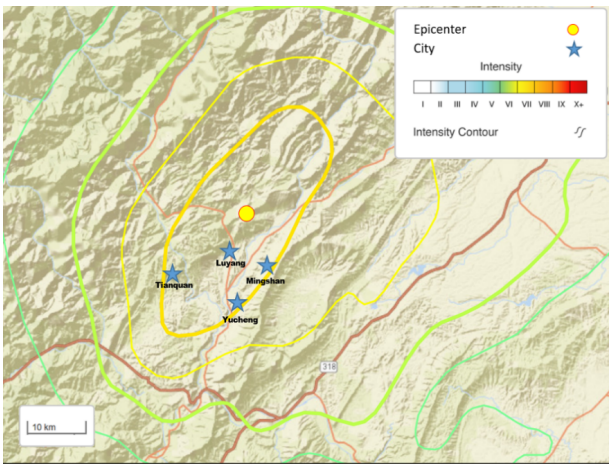


Fig. 6: Areas Affected by the Ya'an Earthquake

case study, we consider 12 areas that were severely attacked by the earthquake within the inner orange circle, including four cities (blue star icons) and eight towns. All 12 of the affected places are potential alternatives for building warehouses during our planning stage. Meanwhile, after an earthquake occurs, demand also comes from the same 12 places. After obtaining geographic information from Google Maps,

we abstractly depict the affected areas as Figure 7. Suppose that the service radius of cities and towns are

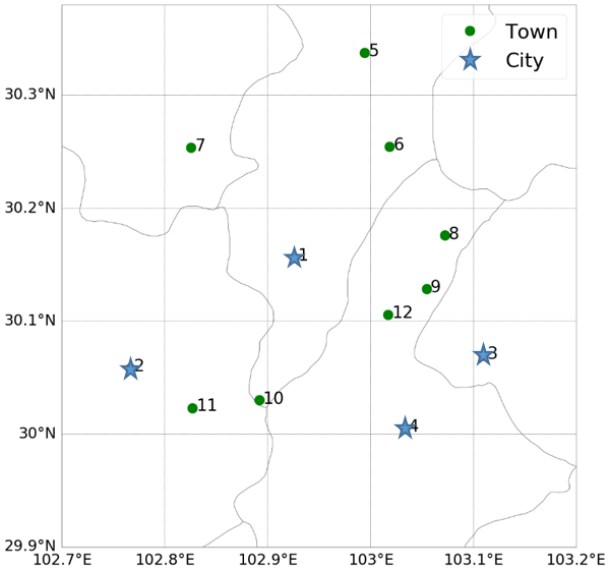


Fig. 7: Areas Affected by the Ya'an Earthquake

30 kilometres and 20 kilometres, respectively. Thus, the initial network structure is

$$\mathcal{E} = \begin{pmatrix} 1&0&0&0&0&1&1&0&0&1&1&0 \\ 0&1&0&0&0&0&0&0&0&1&1&0 \\ 0&0&1&1&0&0&0&0&0&0&0&0 \\ 0&0&1&1&0&0&0&1&1&1&0&1 \\ 0&0&0&0&1&1&0&0&0&0&0&0 \\ 1&0&0&0&1&1&0&0&0&0&0&0 \\ 0&0&0&0&1&0&1&0&0&0&0&0 \\ 0&0&0&0&0&0&0&1&1&0&0&1 \\ 0&0&0&0&0&0&0&1&1&0&0&1 \\ 1&0&0&1&0&0&0&0&0&1&1&0 \\ 0&1&0&0&0&0&0&0&0&1&1&0 \\ 0&0&0&1&0&0&0&1&1&0&0&1 \end{pmatrix}.$$

Note that the 12 places in the case study are potential locations for warehouses and demand nodes. To complete the case study, we collected data about the considered areas from official channels and then calculated model input parameters, such as demand information, the warehouse setup costs, and the holding costs.

To estimate the average relief material package demand from victims, we are required to know the total population in affected areas. Fortunately, the total population information is accessible from the official province-level Statistical Yearbook. Additionally, the news reported[2] that, on average, 16 people lived in one relief tent. Therefore, it is reasonable to believe that each relief package can serve 16 victims. After obtaining these data, without loss of generality, we regard half of the total relief package demand as the mean demand, i.e., $\boldsymbol{\mu}$ =[980, 1231, 3992, 4068, 395, 763, 530, 394, 448, 400, 602, 326]. To calculate the setup cost, we need to consider the type of building materials, price fluctuations due to earthquakes, and the extra cost

[2] http://cpc.people.com.cn/n/2013/0508/c64387-21400852.html

of improving warehouses' resistance to earthquakes. According to the cost calculation formulation in Wang (2010), we have $f = 900 \times 120.53\% \times (1+P) \times M$, where $P \sim U[5\%, 10\%]$ is the additional cost to improve warehouses' resistance, and $M \sim U[3000, 4500]$ is the area of a warehouse. Typically, relief warehouses are made out of brick and concrete, and the price for one square meter is 900 ¥. Furthermore, building material shortages would lead to a 20.53% rise in price. Therefore, normalized by the unit penalty cost of 100,000 ¥, the final setup cost is $\boldsymbol{f} = (51.26, 47.23, 50.47, 48.88, 42.11, 49.27, 44.48, 44.82, 42.96, 35.11, 46.76, 41.80)$. To estimate the holding cost, we first gathered a material list[3] that recorded relief materials delivered to the affected areas within four days, as shown in Table 6. 47,000 tents can be regarded as 47,000 unit pack-

Table 6: Relief Materials Delivered to Ya'an within Four Days

Materials	Quantity	Price (¥)	Shelf Life (Year)
Tent	47,000	3,000	10
Quilt	199,000	399	5
Clothes	10,000	489	5
Folding Bed	10,000	2,000	10
Portable Toilet	200	3,380	5
Water-proof Cloth	700,000	3	5
Candle	100,000	1	10
Instant Noodle	70,000	48	1
Drinking Water	40,000	48	1
Emergency Lamp	900	1,350	5

ages. All prices are collected from online shopping platforms such as Taobao.com or JD.com. Thus, the holding cost for each relief package is 5418 ¥. After incorporating randomness, we obtain the normalized holding cost $\boldsymbol{h} = (0.058, 0.056, 0.056, 0.056, 0.057, 0.053, 0.056, 0.058, 0.059, 0.059, 0.057, 0.053)$. Similarly, we set coefficients of variation of demand nodes as $c_v = 0.8$, while changing correlation coefficient ρ from 0.1 to 0.9 with a step of 0.1 to examine the impact of correlation intensity. Variances of demand $\boldsymbol{\sigma}$ and covariance matrix $\boldsymbol{\Sigma}_{\rho,c_v}$ are calculated by the same procedure that we used in Section 5.2.1. We randomly draw 50 demand samples from a truncated multivariate Gaussian distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma}_{\rho,c_v})$ as input demand samples. Based on these samples, we solve Model (CO), Model (MV), and Model (SAA) to obtain the prepositioning network. Then 1,500 samples, regarded as realized demand, are independently drawn from a mix of distributions consisting of a multivariate Gaussian distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma}_{\rho,c_v})$, a two-point distribution $\mathbb{P}(\tilde{\boldsymbol{d}} = \boldsymbol{\mu} - \boldsymbol{\sigma}) = \mathbb{P}(\tilde{\boldsymbol{d}} = \boldsymbol{\mu} + \boldsymbol{\sigma}) = 0.5$, and a truncated uniform distribution $U\left[\boldsymbol{\mu} - \sqrt{3}\boldsymbol{\sigma}, \boldsymbol{\mu} + \sqrt{3}\boldsymbol{\sigma}\right]$. We calculate their average performances and three different quantile performances, i.e., 90%, 95%, and 99%. Note that quantile performances quantify the model's robustness to hedge the worst-case situation. We exploit different data generation processes when generating input samples and test samples as conducted in Ni et al. (2018) to mimic the reality that a wrong distribution assumption is a possibility.

Table 7 reveals the designed prepositioning networks when $\rho = 0.2, 0.5, 0.8$. It is easy to see that in all cases, node 4 and node 6 are selected to build warehouses partially due to their high connectivities. Moreover, as ρ increases, the total inventories of Model (SAA) and Model (CO) increase, but that of Model (MV) does not change substantially.

Figure 8 displays the relationship between ρ and fulfilment rates on average and under different VaR percentiles. The first subfigure in Figure 8 shows that, on average, Model (SAA) and Model (CO) can achieve an almost 100% fulfilment rate while that of Model (MV) is slightly lower. In cases of 90%, 95%, and 99% VaR, Model (CO) can still achieve the highest fulfilment rate, but the other two models perform much worse, as shown in the other three subfigures. Notably, the fulfilment rate of Model (MV) decreases

[3] http://www.mca.gov.cn/article/zwgk/mzyw/201304/20130400448512.shtml, available on May, 2020

Table 7: Networks Designed for the Case Study

ρ	Model	Node	1	2	3	4	5	6	7	8	9	10	11	12	Total Inv.
0.2	SAA		1403	2976		15321		2057				2206			23963
	CO		2680	3687		18223		2701							27291
	MV		2789			17095		2111					3243		25238
0.5	SAA		2037	2990		18006		3245							26278
	CO			4265		19497		4229	1148						29139
	MV		2693	3154		17074		2031							24952
0.8	SAA		1920	3574		18643		4053							28190
	CO			4424		21157		4923	1209						31713
	MV		2828			16980		2230					3328		25366

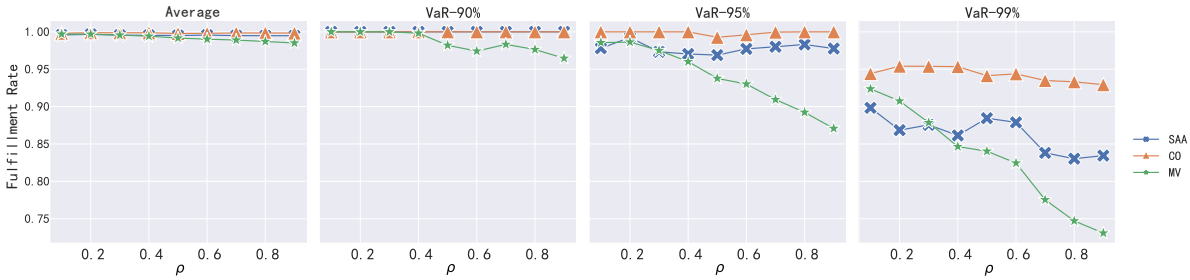


Fig. 8: Case Study: Fulfilment Rate Comparison

as ρ increases since it is limited in considering demand correlations. Figure 9 expresses the total costs

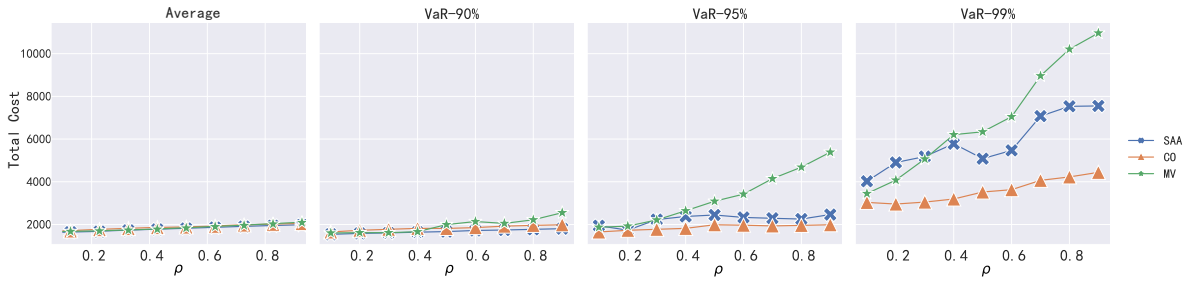


Fig. 9: Case Study: Total Cost Comparison

of the models. Suprisingly, the total costs of the three models are almost the same on average. However, in case of 90%, 95%, and 99% VaR, Model (SAA) and Model (MV) result in much higher total costs compared to Model (CO). It demonstrates Model (CO)'s capability of addressing extreme cases. Moreover, the increasing curve of Model (CO) always remain flat, while the curves of the other two models surge rapidly when ρ increases, which verifies that Model (CO) can accurately capture and correctly handle the correlated demand uncertainty while the other models cannot.

In summary, Figure 8 and 9 show that our model can provide a higher fulfilment rate with a lower total cost in the worse case situation. Furthermore, the robustness of our model validate that the introduction of demand correlation in network design is necessary.

6 Extensions

6.1 Multiple materials

Suppose there are K different kinds of relief materials, and for each relief material $k \in [k]$, the first moment $\boldsymbol{\mu}_k$ and second moment $\boldsymbol{\Sigma}_k$ are known to decision makers. Further, assume that the demand variables are uncorrelated over relief materials. Under these assumptions, the second-stage problem is separable and can be modified as:

$$\begin{aligned} \min_{\boldsymbol{y}_k, \boldsymbol{z}} \ & \kappa(\boldsymbol{f}^T \boldsymbol{z} + \sum_{k \in [K]} \boldsymbol{h}_k^T \boldsymbol{y}_k) + \sum_{k \in [K]} \sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)} \mathbb{E}_{\mathbb{P}} \left[g_k(\boldsymbol{y}_k, \tilde{\boldsymbol{d}}) \right] \\ \text{s.t.} \ & \boldsymbol{y}_k \leq U \boldsymbol{z}, \ \forall k \in [K]; \\ & \boldsymbol{y}_k \in \mathbb{R}_+^m, \boldsymbol{z} \in \{0,1\}^m. \end{aligned} \tag{10}$$

Let $L_k(\boldsymbol{y}) = \sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)} \mathbb{E}_{\mathbb{P}} \left[g_k(\boldsymbol{y}, \tilde{\boldsymbol{d}}) \right]$. Following the analysis in Section 4, we can reformulate $L_k(\boldsymbol{y})$ as a conic programming problem. The resulting model contains $|K|$ copositive cone constraints, which slows the computational process, but theoretically, with positive semidefinite relaxation, it does not make the model intractable.

6.2 Capacitated Warehouses

Capacitated warehouses are common in reality, and the setup cost of warehouses $\boldsymbol{f}$ can be capacity-specific. Since these constraints are imposed in the first stage, they are easy to incorporate into our model. Suppose L types of warehouses are available, each of which has a limited capacity $C_l, \forall l \in [L]$ with a fixed setup cost $\boldsymbol{f}_l, \forall l \in [L]$. Now, we can revise the reduced model (rDRND) as:

$$\begin{aligned} \min_{\boldsymbol{y}, \boldsymbol{z}_l} \ & \kappa(\sum_{l \in [L]} \boldsymbol{f}_l^T \boldsymbol{z}_l + \boldsymbol{h}^T \boldsymbol{y}) + \sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu}, \boldsymbol{\Sigma})} \mathbb{E}_{\mathbb{P}} \left[g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) \right] \\ \text{s.t.} \ & \boldsymbol{y} \leq \sum_{l \in [L]} C_k \boldsymbol{z}_l, \ \forall l \in [L]; \\ & \sum_{l \in [L]} \boldsymbol{z}_l \leq \mathbf{1}; \\ & \boldsymbol{y} \in \mathbb{R}_+^m, \boldsymbol{z}_l \in \{0,1\}^m, \forall l \in [L]. \end{aligned} \tag{11}$$

The objective function is self-evident. The first constraint requires that the number of stored products does not exceed warehouse capacity limits, while the second constraint restricts the number of warehouses at each location to one. We can still follow the same analysis as in Section 4 to reformulate the *sup* problem, and the resulting conic model is still tractable under relaxation, although it involves additional binary variables. One notable result is that imposing additional tractable constraints on the first-stage problem does not theoretically increase the difficulty of obtaining the solution. Therefore, our proposed model is compatible with additional constraints in the first-stage problem, as long as they are polynomial-time-solvable constraints, such as linear constraints, second-order cone constraints, and positive semidefinite constraints.

6.3 Uncertainty in the First and Second Moments

Data-driven approaches have been an emerging direction in the operations management community, and our model can perfectly fit such an approach. Given the basis of historical demand samples, the estimated first and second moment could be inaccurate, and it is necessary to capture the estimation error during modelling. We introduce the estimation error as uncertainties; namely, the true first and second moment are subjected to an uncertainty set constructed from the estimated moments. We then utilize the technique proposed in Delage and Ye (2010). Suppose that the ground-truth first moment $\boldsymbol{\mu}$ and second cross-moment $\boldsymbol{\Sigma}$ are uncertain and that the uncertainty set is defined as:

$$(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \in \mathscr{D} := \left\{ (\boldsymbol{\mu}, \boldsymbol{\Sigma}) \,\middle|\, \begin{array}{l} (\boldsymbol{\mu} - \widehat{\boldsymbol{\mu}})^T \boldsymbol{\Sigma}_0^{-1} (\boldsymbol{\mu} - \widehat{\boldsymbol{\mu}}) \leq \gamma_1 \\ \boldsymbol{\Sigma} \preceq \gamma_2 \widehat{\boldsymbol{\Sigma}} + \boldsymbol{\mu}\widehat{\boldsymbol{\mu}}^T + \widehat{\boldsymbol{\mu}}\boldsymbol{\mu}^T - \widehat{\boldsymbol{\mu}}\widehat{\boldsymbol{\mu}}^T \end{array} \right\},$$

where $\widehat{\boldsymbol{\mu}}$ and $\widehat{\boldsymbol{\Sigma}}$ are the estimated first and second moment describing an ellipsoid. Parameters $\gamma_1 \geq 0$ and $\gamma_2 \geq 1$ naturally quantify one's confidence in $\widehat{\boldsymbol{\mu}}$ and $\widehat{\boldsymbol{\Sigma}}$. Therefore, incorporating new decision variables $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ into Model (rDRND) yields the following model:

$$\begin{aligned} \min_{\boldsymbol{y},\boldsymbol{z}} \ & \kappa(\boldsymbol{f}^T \boldsymbol{z} + \boldsymbol{h}^T \boldsymbol{y}) + \max_{(\boldsymbol{\mu},\boldsymbol{\Sigma}) \in \mathscr{D}} \sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu},\boldsymbol{\Sigma})} \mathbb{E}_{\mathbb{P}} \left[g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) \right] \\ \text{s.t.} \ & \boldsymbol{y} \leq U\boldsymbol{z}; \\ & \boldsymbol{y} \in \mathbb{R}_+^m, \boldsymbol{z} \in \{0,1\}^m. \end{aligned} \tag{12}$$

Recall that $\sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu},\boldsymbol{\Sigma})} \mathbb{E}_{\mathbb{P}} \left[g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) \right]$ can be reformulated as a completely positive programming maximization problem, which can be combined with the decision $(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \in \mathscr{D}$. Since $\mathscr{D}$ involves only positive semidefinite constraints, the strong duality still holds between the copositive programming minimization problem and the completely positive programming maximization problem. The remaining analysis is the same.

7 Conclusion

Motivated by the existence of spatially correlated demand uncertainty in disaster relief management, we studied a disaster relief network design problem under demand uncertainty. The demand uncertainty is described as an ambiguity set characterized by the first two moments. This problem is formulated as a mixed-integer two-stage distributionally robust optimization problem, with the objective to minimize the setup cost, holding cost, and the penalty cost due to unmet demand. Note that minimizing the penalty cost of unmet demand implies that our objective is to maximize the total demand fulfilment, which is one of the most critical measures in disaster management. We proposed an equivalent reformulation to transform the model into a mixed-integer conic problem based on copositive cones. We also proposed a pricing policy with warm starts to accelerate the branch-and-pricing process. Extensive numerical studies on synthetic datasets and a case study on the Ya'an Earthquake demonstrate the superiority of our proposed model and the necessity of incorporating spatially correlated demand uncertainty when it exists. Meanwhile, sensitivity analysis shows that stronger correlations would amplify the improvement, further stressing the necessity of considering demand correlations. Several extensions have been proposed to express the compatibility of our proposed model by involving more realistic features.

Our study also demonstrated the superiority of copositive programming and its advantages in solving operations management problems. We hope that our study will motivate further research on copositive programming and its applications in the field of operations management. One potential research direction is to consider a more realistic recourse problem that includes shipping costs, ordering costs, location-specified penalty costs, etc. Interested readers are also encouraged to explore the analytic properties of copositive cones mathematically.

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A Proofs

A.1 Proof for Proposition 1

Proof We prove this proposition by showing $g(\boldsymbol{y}, \boldsymbol{z}, \tilde{\boldsymbol{d}}) \geq g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ and $g(\boldsymbol{y}, \boldsymbol{z}, \tilde{\boldsymbol{d}}) \leq g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ both hold. For clarity, we rewrite both problems as follows:

$$
\begin{aligned}
g(\boldsymbol{y}, \boldsymbol{z}, \tilde{\boldsymbol{d}}) = \min_{\boldsymbol{t}} \quad & \sum_{j \in [n]} \tilde{d}_j - \sum_{(i,j) \in \mathcal{E}(\boldsymbol{z})} t_{ij} \\
s.t. \quad & \sum_{i:(i,j) \in \mathcal{E}(\boldsymbol{z})} t_{ij} \leq \tilde{d}_j, \forall j \in [n] \\
& \sum_{j:(i,j) \in \mathcal{E}(\boldsymbol{z})} t_{ij} \leq y_i, \forall i \in [m] \\
& t_{ij} \geq 0, \forall (i,j) \in \mathcal{E}(\boldsymbol{z})
\end{aligned} \tag{13}
$$

Define an index set $\boldsymbol{z}^+$ containing all location indices for building a warehouse, i.e., $\boldsymbol{z}^+ := \{i \,|z_i = 1, \ \forall i \in [m]\}$.

$$
\begin{aligned}
g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) = \min_{\boldsymbol{t}} \quad & \sum_{j \in [n]} \tilde{d}_j - \sum_{(i,j) \in \mathcal{E}(\boldsymbol{z})} t_{ij} - \sum_{(i,j) \notin \mathcal{E}(\boldsymbol{z})} t_{ij} \qquad (14) \\
s.t. \quad & \sum_{i:(i,j) \in \mathcal{E}(\boldsymbol{z})} t_{ij} + \sum_{i:(i,j) \notin \mathcal{E}(\boldsymbol{z})} t_{ij} \leq \tilde{d}_j, \forall j \in [n] \\
& \sum_{j:(i,j) \in \mathcal{E}} t_{ij} \leq y_i, \forall i \in \boldsymbol{z}^+ \\
& \sum_{j:(i,j) \in \mathcal{E}} t_{ij} \leq 0, \forall i \notin \boldsymbol{z}^+ \qquad (15) \\
& t_{ij} \geq 0, \forall (i,j) \in \mathcal{E}
\end{aligned}
$$

In $g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$, the right-hand-side coefficient of the third constraint is zero since establishing a warehouse is a prerequisite for storing products.

1. $g(\boldsymbol{y}, \boldsymbol{z}, \tilde{\boldsymbol{d}}) \geq g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$
   Clearly, the optimal solution $\boldsymbol{t}^*$ to (13) is always a feasible solution to (14) by setting other $t_{ij} = 0, (i,j) \notin \mathcal{E}(\boldsymbol{z})$. Therefore, $g(\boldsymbol{y}, \boldsymbol{z}, \tilde{\boldsymbol{d}}) \geq g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ holds.
2. $g(\boldsymbol{y}, \boldsymbol{z}, \tilde{\boldsymbol{d}}) \leq g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$
   Now, suppose $\boldsymbol{t}^*$ is the optimal solution to (14). We split the solution into two parts according to whether $i \in \boldsymbol{z}^+$, i.e., $\boldsymbol{t}^* = (\boldsymbol{t}^+, \boldsymbol{t}^0)$, where $\boldsymbol{t}^+$ contains the allocation decision from warehouses and $\boldsymbol{t}^0$ contains the allocation decisions from locations where no warehouse is built. From constraint (15), we have $\boldsymbol{t}^0 = \boldsymbol{0}$. It is easy to check that $\boldsymbol{t}^+$ is a feasible solution to (13). Therefore, $g(\boldsymbol{y}, \boldsymbol{z}, \tilde{\boldsymbol{d}}) \leq g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ always holds.

Combining the above two statements completes the proof.

A.2 Proof for Proposition 2

$$
\begin{aligned}
g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) = & \min_{\boldsymbol{t} \in \Omega(\boldsymbol{y})} \sum_{j \in [n]} \tilde{d}_j - \sum_{(i,j) \in \mathcal{E}} t_{ij} \\
= & \sum_{j \in [n]} \tilde{d}_j - \max_{\boldsymbol{t} \in \Omega(\boldsymbol{y})} \sum_{(i,j) \in \mathcal{E}} t_{ij} \\
= & \sum_{j \in [n]} \tilde{d}_j - \min_{(\boldsymbol{u}, \boldsymbol{v}) \in \mathcal{L}} \tilde{\boldsymbol{d}}^T \boldsymbol{u} + \boldsymbol{y}^T \boldsymbol{v} \\
= & - \min_{(\boldsymbol{u}, \boldsymbol{v}) \in \mathcal{L}} \tilde{\boldsymbol{d}}^T (\boldsymbol{u} - \boldsymbol{1}) + \boldsymbol{y}^T \boldsymbol{v} \\
= & \max_{(\boldsymbol{u}, \boldsymbol{v}) \in \mathcal{L}} \tilde{\boldsymbol{d}}^T (\boldsymbol{1} - \boldsymbol{u}) - \boldsymbol{y}^T \boldsymbol{v}.
\end{aligned}
$$

The first equation is the unmet demand minimization problem after $\tilde{\boldsymbol{d}}$ is realized. The second equation is obtained by simply exchanging the minimizing operator and negative symbol. The third equation strictly follows the classic min-cut max-flow theorem (see Simchi-Levi and Wei (2015)), where the feasible region is

$$\mathcal{L} := \{(\boldsymbol{u}, \boldsymbol{v}) \in \{0,1\}^{n+m} | u_j + v_i \geq 1, \forall (i,j) \in \mathcal{E}\}.$$

The fourth equation holds by combining the first term and the minimization problem in the third equation, where $\mathbf{1}$ is a vector with all elements equal to 1. The last equation comes from exchanging the maximizing operator and the negative symbol. Additionally, it is obvious that the set $\mathcal{L}$ is always nonempty as long as $\mathcal{E}$ forms a bipartite graph without isolated nodes..

For further analysis, we modify $\mathcal{L}$ by replacing $\mathbf{1} - \boldsymbol{u}$ with decision variables $\hat{\boldsymbol{u}} = \mathbf{1} - \boldsymbol{u}$ and define the new region with $\hat{\boldsymbol{u}}$ as

$$\hat{\mathcal{L}} := \{(\hat{\boldsymbol{u}}, \boldsymbol{v}) \in \{0,1\}^{n+m} | v_i \geq \hat{u}_j, \forall (i,j) \in \mathcal{E}\}.$$

Moreover, the linear relaxation of the feasible region is

$$\hat{\mathcal{L}}_{lp} := \left\{ (\hat{\boldsymbol{u}}, \boldsymbol{v}) \in \mathbb{R}_+^{n+m} \left| \begin{array}{l} v_i \geq \hat{u}_j, \forall (i,j) \in \mathcal{E} \\ \hat{\boldsymbol{u}} \leq \mathbf{1} \\ \boldsymbol{v} \leq \mathbf{1} \end{array} \right. \right\}.$$

Therefore, $g(\boldsymbol{y}, \tilde{\boldsymbol{d}})$ is equivalent among the following formulations:

$$\begin{aligned} g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) &= \max_{(\boldsymbol{u},\boldsymbol{v}) \in \mathcal{L}} \tilde{\boldsymbol{d}}^T (\mathbf{1} - \boldsymbol{u}) - \boldsymbol{y}^T \boldsymbol{v} \\ &= \max_{(\hat{\boldsymbol{u}},\boldsymbol{v}) \in \hat{\mathcal{L}}} \tilde{\boldsymbol{d}}^T \hat{\boldsymbol{u}} - \boldsymbol{y}^T \boldsymbol{v} \\ &= \max_{(\hat{\boldsymbol{u}},\boldsymbol{v}) \in \hat{\mathcal{L}}_{lp} \cap \{0,1\}^{n+m}} \tilde{\boldsymbol{d}}^T \hat{\boldsymbol{u}} - \boldsymbol{y}^T \boldsymbol{v}. \end{aligned}$$

A.3 Proof for Proposition 3

Proof Instead of considering $g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) = \max_{(\hat{\boldsymbol{u}},\boldsymbol{v}) \in \hat{\mathcal{L}}_{lp} \cap \{0,1\}^{n+m}} \tilde{\boldsymbol{d}}^T \hat{\boldsymbol{u}} - \boldsymbol{y}^T \boldsymbol{v}$, we consider its linear relaxation problem $\max_{(\hat{\boldsymbol{u}},\boldsymbol{v}) \in \hat{\mathcal{L}}_{lp}} \tilde{\boldsymbol{d}}^T \hat{\boldsymbol{u}} - \boldsymbol{y}^T \boldsymbol{v}$. We prove this proposition by demonstrating that problem (5) with the linear relaxation problem is NP-hard according to **?**. As discretizing the feasible region does not reduce the computational complexity, (5) is at least NP-hard.

First, we take the dual on (5) as follows:

$$\begin{aligned} L(\boldsymbol{y}) = \sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu}, \boldsymbol{\Sigma})} \mathbb{E}_{\mathbb{P}}[g(\boldsymbol{y}, \tilde{\boldsymbol{d}})] = \quad & \sup_{m(\tilde{\boldsymbol{d}})} \int_{\mathbb{R}_+^n} g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) \; m(\tilde{\boldsymbol{d}}) \\ s.t. & \int_{\mathbb{R}_+^n} 1 \; m(\tilde{\boldsymbol{d}}) = 1 \\ & \int_{\mathbb{R}_+^n} \tilde{\boldsymbol{d}} \; m(\tilde{\boldsymbol{d}}) = \boldsymbol{\mu} \\ & \int_{\mathbb{R}_+^n} \tilde{\boldsymbol{d}} \tilde{\boldsymbol{d}}^T \; m(\tilde{\boldsymbol{d}}) = \boldsymbol{\Sigma} \\ & m(\tilde{\boldsymbol{d}}) \in \mathscr{M}_+(\mathbb{R}_+^n) \end{aligned}$$

where $m(\tilde{\boldsymbol{d}})$ is a probability measure on $\mathbb{R}_+^n$. The dual problem is:

$$\begin{aligned} & \inf_{s, \boldsymbol{t}, \boldsymbol{\Xi}} s + \boldsymbol{\mu}^T \boldsymbol{t} + \boldsymbol{\Sigma} \bullet \boldsymbol{\Xi} \\ & s.t. \quad \min_{(\hat{\boldsymbol{u}},\boldsymbol{v}) \in \hat{\mathcal{L}}_{lp}, \tilde{\boldsymbol{d}} \in \mathbb{R}_+^n} \tilde{\boldsymbol{d}} \boldsymbol{\Xi} \tilde{\boldsymbol{d}} + \tilde{\boldsymbol{d}}^T (\boldsymbol{t} - \hat{\boldsymbol{u}}) + \boldsymbol{y}^T \boldsymbol{v} + s \geq 0 \end{aligned} \tag{16}$$

Clearly, strong duality holds between the maximization problem and the dual minimization problem. According to Theorem 3.1 in **?**, the separation problem, i.e., for a given $(s, \boldsymbol{t}, \boldsymbol{\Xi})$, check if the constraint (16) is satisfied and if not, find a feasible $(\hat{\boldsymbol{u}}, \boldsymbol{v}) \in \hat{\mathcal{L}}_{lp}, \tilde{\boldsymbol{d}} \in \mathbb{R}_+^n$ satisfying (16), is NP-complete. Because of the equivalence of separation and optimization, the minimization problem is NP-hard. Therefore, evaluating the value of $L(\boldsymbol{y})$ is at least NP-hard.

A.4 Proof for Proposition 4

Proof Before presenting the detailed proof, we concisely summarize the roadmap to prove this proposition. We first transform the second-stage problem $L(\boldsymbol{y})$ into a completely positive conic maximization problem according to Theorem 3.3 in Natarajan et al. (2011). Then, we take the dual to obtain a minimization problem on the copositive cone. We further show that strong duality holds between the maximization and the minimization problems. Finally, we combine the second-stage minimization problem with the first-stage problem to obtain (CO).

Step 1: Transform $L(\boldsymbol{y})$

Let $M = m + n + r$ and $N = 2m + 2n + r$. By adding slackness variables $(\boldsymbol{s}^\dagger, \hat{\boldsymbol{u}}^\dagger, \boldsymbol{v}^\dagger) \in \mathbb{R}_+^M$, we can transform $\hat{\mathcal{L}}_{lp}$ into a system that consists of only equalities:

$$\hat{\mathcal{L}}_{lp}^\dagger := \left\{ (\hat{\boldsymbol{u}}, \boldsymbol{v}, \boldsymbol{s}^\dagger, \hat{\boldsymbol{u}}^\dagger, \boldsymbol{v}^\dagger) \in \mathbb{R}_+^N \,\middle|\, \begin{array}{l} \hat{u}_j - v_i + s_{ij}^\dagger = 0, \forall (i,j) \in \mathcal{E} \\ \hat{\boldsymbol{u}} + \hat{\boldsymbol{u}}^\dagger = \mathbf{1} \\ \boldsymbol{v} + \boldsymbol{v}^\dagger = \mathbf{1} \end{array} \right\}.$$

Therefore, the expected second-stage penalty under the worst-case distribution $L(\boldsymbol{y})$ possesses following relations:

$$L(\boldsymbol{y}) = \sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu}, \boldsymbol{\Sigma})} \mathbb{E}_{\mathbb{P}} \left[g(\boldsymbol{y}, \tilde{\boldsymbol{d}}) \right] = \sup_{\mathbb{P} \in \mathscr{F}(\boldsymbol{\mu}, \boldsymbol{\Sigma})} \mathbb{E}_{\mathbb{P}} \left[\max_{(\hat{\boldsymbol{u}}, \boldsymbol{v}, \boldsymbol{s}^\dagger, \hat{\boldsymbol{u}}^\dagger, \boldsymbol{v}^\dagger) \in \hat{\mathcal{L}}_{lp}^\dagger \cap \{0,1\}^N} \tilde{\boldsymbol{d}}^T \hat{\boldsymbol{u}} - \boldsymbol{y}^T \boldsymbol{v} \right].$$

Note that the uncertainties of the inner problem exist in only the objective function. For simplicity, we denote the decision variables by $\boldsymbol{x} := (\hat{\boldsymbol{u}}, \boldsymbol{v}, \boldsymbol{s}^\dagger, \hat{\boldsymbol{u}}^\dagger, \boldsymbol{v}^\dagger) \in \hat{\mathcal{L}}_{lp}^\dagger \subset \mathbb{R}_+^N$. The constraints of $\hat{\mathcal{L}}_{lp}^\dagger$ can also be written in a more general form as $\left\{ \boldsymbol{x} \geq \mathbf{0} \mid \boldsymbol{a}_i^T \boldsymbol{x} = b_i, \forall i \in [M] \right\}$. It is easy to check the following two statements hold:

1. $\boldsymbol{x} \in \hat{\mathcal{L}}_{lp}^\dagger := \left\{ \boldsymbol{x} \geq \mathbf{0} \,\middle|\, \boldsymbol{a}_i^T \boldsymbol{x} = \boldsymbol{b}_i, \forall i \in [M] \right\} \implies \boldsymbol{x} \leq \mathbf{1}$;
2. The feasible region of $\hat{\mathcal{L}}_{lp}^\dagger \cap \{0,1\}^N$ is nonempty and bounded.

We then define three new variables.

- $\boldsymbol{p} := \mathbb{E}_{\mathbb{P}}[\boldsymbol{x}(\tilde{\boldsymbol{d}})] \in \mathbb{R}_+^N$
- $\boldsymbol{Y} := \mathbb{E}_{\mathbb{P}}[\boldsymbol{x}(\tilde{\boldsymbol{d}})\tilde{\boldsymbol{d}}^T] \in \mathbb{R}_+^{N \times n}$
- $\boldsymbol{X} := \mathbb{E}_{\mathbb{P}}[\boldsymbol{x}(\tilde{\boldsymbol{d}})\boldsymbol{x}(\tilde{\boldsymbol{d}})^T] \in \mathbb{R}_+^{N \times N}$

According to Theorem 3.3 in Natarajan et al. (2011), we can equivalently reformulate $L(\boldsymbol{y})$ to a completely positive program $L_{CP}(\boldsymbol{y})$ as follows:

$$\begin{aligned} L(\boldsymbol{y}) = L_{CP}(\boldsymbol{y}) = \max_{\boldsymbol{p}, \boldsymbol{Y}, \boldsymbol{X}} & \begin{bmatrix} \boldsymbol{E}^{[n]} \\ \mathbf{0} \end{bmatrix} \bullet \boldsymbol{Y} + \begin{bmatrix} \mathbf{0}_{n \times 1} \\ -\boldsymbol{y}_{m \times 1} \\ \mathbf{0}_{M \times 1} \end{bmatrix}^T \boldsymbol{p} \\ s.t. \quad & \boldsymbol{a}_i^T \boldsymbol{p} = b_i, \forall i \in [M]; && (17) \\ & (\boldsymbol{a}_i \boldsymbol{a}_i^T) \bullet \boldsymbol{X} = b_i^2, \forall i \in [M]; && (18) \\ & \boldsymbol{E}_{(j,j)} \bullet \boldsymbol{X} - \boldsymbol{e}_{(j)}^T \boldsymbol{p} = 0, \forall j \in [N]; && (19) \\ & \begin{pmatrix} 1 & \boldsymbol{\mu}^T & \boldsymbol{p}^T \\ \boldsymbol{\mu} & \boldsymbol{\Sigma} & \boldsymbol{Y}^T \\ \boldsymbol{p} & \boldsymbol{Y} & \boldsymbol{X} \end{pmatrix} \succeq_{cp} \mathbf{0}, && (20) \end{aligned}$$

where $\boldsymbol{p} \in \mathbb{R}_+^N$, $\boldsymbol{Y} \in \mathbb{R}_+^{N \times n}$, and $\boldsymbol{X} \in \mathbb{R}_+^{N \times N}$ are decision variables. We use superscript $^{[k]}$ to indicate a k-by-k matrix variable. $\boldsymbol{E}^{[n]}$ is an n-dimensional identity matrix. Additionally, we use $\boldsymbol{E}_{(j,j)}$ to represent a square matrix with the (j,j)-position element equal to 1 and others as 0. Furthermore, $\boldsymbol{e}_{(j)}$ is the unit vector with the j-th element equal to 1. $\succeq_{cp}$ means that the matrix belongs to a completely positive cone. Unless stated otherwise, $\mathbf{0}$ is a zero vector or zero matrix with the appropriate shape.

Step 2: Take duality on $L_{CP}(\boldsymbol{y})$

Taking the duality on $L_{CP}(\boldsymbol{y})$ yields the following results

$$L_{CO}(\boldsymbol{y}) = \min_{\alpha_i,\beta_i,\theta_j,\tau,\boldsymbol{\xi},\boldsymbol{\varphi},\boldsymbol{\Psi},\boldsymbol{\Xi},\boldsymbol{W}} \sum_{i\in[M]} b_i\alpha_i + b_i^2\beta_i + \tau + 2\boldsymbol{\mu}^T\boldsymbol{\xi} + \boldsymbol{\Sigma}\bullet\boldsymbol{\Xi}$$

$$s.t. \sum_{i\in[M]} \boldsymbol{a}_i\alpha_i - \sum_{j\in[N]} \boldsymbol{e}_{(j)}\theta_j - 2\boldsymbol{\varphi} = \begin{bmatrix} \boldsymbol{0}_{n\times 1} \\ -\boldsymbol{y}_{m\times 1} \\ \boldsymbol{0}_{M\times 1} \end{bmatrix}; \tag{21}$$

$$\sum_{i\in[M]} \boldsymbol{a}_i\boldsymbol{a}_i^T\beta_i + \sum_{j\in[N]} \boldsymbol{E}_{(j,j)}\theta_j - \boldsymbol{W} = \boldsymbol{0}; \tag{22}$$

$$-2\boldsymbol{\Psi} = \begin{bmatrix} \boldsymbol{E}^{[n]} \\ \boldsymbol{0} \end{bmatrix}; \tag{23}$$

$$\begin{pmatrix} \tau & \boldsymbol{\xi}^T & \boldsymbol{\varphi}^T \\ \boldsymbol{\xi} & \boldsymbol{\Xi} & \boldsymbol{\Psi}^T \\ \boldsymbol{\varphi} & \boldsymbol{\Psi} & \boldsymbol{W} \end{pmatrix} \succeq_{co} \boldsymbol{0}. \tag{24}$$

α_i, β_i, and θ_j are the dual variables for constraint (17), (18), and (19), respectively. $\begin{pmatrix} \tau & \boldsymbol{\xi}^T & \boldsymbol{\varphi}^T \\ \boldsymbol{\xi} & \boldsymbol{\Xi} & \boldsymbol{\Psi}^T \\ \boldsymbol{\varphi} & \boldsymbol{\Psi} & \boldsymbol{W} \end{pmatrix}$ is the dual variable for the completely positive cone constraint (20). According to the weak duality theorem, $L_{CO}(\boldsymbol{y}) \geq L_{CP}(\boldsymbol{y})$.

Step 3: Strong duality holds, i.e. $L_{CO}(\boldsymbol{y}) = L_{CP}(\boldsymbol{y})$

We first divide the solution to (3) into two parts. One is composed of decision variables $\boldsymbol{\theta}$, and the other consists of all slack variables $\boldsymbol{s}$, i.e., $\boldsymbol{x} = \binom{\boldsymbol{\theta}}{\boldsymbol{s}}$, where $\boldsymbol{\theta} = (\hat{\boldsymbol{u}}, \boldsymbol{v}) \in \mathbb{R}_+^{(n+m)}$, $\boldsymbol{s} = (\boldsymbol{s}^\dagger, \hat{\boldsymbol{u}}^\dagger, \boldsymbol{v}^\dagger) \in \mathbb{R}_+^{(r+n+m)}$. According to Theorem 2 in Yan et al. (2018), $L_{CO}(\boldsymbol{y}) = L_{CP}(\boldsymbol{y})$ holds as long as the following two conditions hold:

1. The moment matrix $\begin{pmatrix} 1 & \boldsymbol{\mu}^T \\ \boldsymbol{\mu} & \boldsymbol{\Sigma} \end{pmatrix}$ lies in the interior of a $(1+n)\times(1+n)$-dimensional completely positive cone.
2. There exists a set of feasible solutions $\boldsymbol{x}^{(k)} = \binom{\boldsymbol{\theta}^{(k)}}{\boldsymbol{s}^{(k)}}$, $\forall k \in [K]$ to $\hat{\mathcal{L}}_{lp}^\dagger$ such that $\text{span}\{\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(K)}\} = \mathbb{R}^{n+m}$, and at least one of them is strictly positive; i.e., $\exists \boldsymbol{\theta}^{(l)} \in \{\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(K)}\}$ such that $\boldsymbol{\theta}^{(l)} > \boldsymbol{0}$

Since the first condition is given in the proposition, we only need to prove that the second condition is always satisfied. It is easy to check that $\boldsymbol{x}^{(0)} = \binom{\boldsymbol{\theta}^{(0)}}{\boldsymbol{s}^{(0)}} = \binom{\boldsymbol{1}}{\boldsymbol{0}}$ is a feasible solution, and $\boldsymbol{\theta}^{(0)}$ is strictly positive.

Another two sets of feasible solutions exist:

$$\mathcal{S}_1 := \left\{ \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{s} \end{bmatrix} \in \mathbb{R}_+^N \;\middle|\; \begin{bmatrix} \boldsymbol{\theta}^{(i)} \\ \boldsymbol{s}^{(i)} \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} \boldsymbol{0} \\ \boldsymbol{e}_{(i)} \end{pmatrix} \\ \begin{pmatrix} \sum_{\gamma\in idx(i)} \boldsymbol{e}_{(\gamma)} \\ \boldsymbol{1} \\ \boldsymbol{1} - \boldsymbol{e}_{(i)} \end{pmatrix} \end{bmatrix}, \ \forall i \in [m] \right\},$$

where $idx(i)$ is the index set that contains all indices of roads that are connected to node i, ensuring the first constraint in $\hat{\mathcal{L}}_{lp}^\dagger$ is met. Another set of feasible solutions is:

$$\mathcal{S}_2 := \left\{ \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{s} \end{bmatrix} \in \mathbb{R}_+^N \;\middle|\; \begin{bmatrix} \boldsymbol{\theta}^{(j)} \\ \boldsymbol{s}^{(j)} \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} \boldsymbol{e}_{(j)} \\ \sum_{i\in\Gamma(j)} \boldsymbol{e}_{(i)} \end{pmatrix} \\ \begin{pmatrix} \boldsymbol{0} \\ \boldsymbol{1} - \boldsymbol{e}_{(j)} \\ \boldsymbol{1} - \sum_{i\in\Gamma(j)} \boldsymbol{e}_{(i)} \end{pmatrix} \end{bmatrix}, \ \forall j \in [n] \right\}.$$

It is also easy to check that all vectors $\boldsymbol{\theta}$ in $\mathcal{S}_1 \cup \mathcal{S}_2$ span the space $\mathbb{R}^{(n+m)}$ by Gaussian elimination. Therefore, $\mathcal{S}_1 \cup \mathcal{S}_2 \cup \{\boldsymbol{x}^{(0)}\}$ forms a satisfied solution set that meets the second condition, which completes the current step of the proof.

Step 4: Combine with the first-stage problem

We further combine $L_{CO}(\boldsymbol{y})$ with the first-stage problem and replace some of the decision variables in (24) with equations (21), (22), and (23). Now, we obtain the (rDRND) problem in mixed-integer copositive cone form, i.e., problem (CO).

B Simulations on General Networks

We conduct more simulation experiments on general networks and compare the results with those for $(8, 8)$ networks. Specifically, we consider $(m, n) = (4, 8)$ and $(m, n) = (8, 12)$. Except for changes in the number of potential supply nodes and demand nodes, the other parameters remain the same. We also modify three parameters, namely, the correlation parameter ρ, coefficient of variation c_v, and risk attitude parameter κ, to conduct sensitivity analyses. Generally, we repeat all the experiments in this section but on different networks. Table 8 summarizes the performance of different networks in terms of fulfilment rate (FR). We regard the performance of Model (SAA) as the benchmark, and the ratios in Table 8 are calculated according to $\frac{FR_x - FR_{SAA}}{FR_{SAA}} \times 100\%, x \in \{CO, MV\}$.

Table 8: Fulfilment Rate Comparison of General Networks

Fulfilment Rate			MGD									MixD								
Relative Changes			(4, 8)			(8, 8)			(8, 12)			(4, 8)			(8, 8)			(8, 12)		
ρ	c_v	κ	SAA	CO	MV	SAA	CO	MV	SAA	CO	MV	SAA	CO	MV	SAA	CO	MV	SAA	CO	MV
0.05	0.3	1.0	-	0.30%	0.12%	-	0.35%	0.21%	-	0.31%	0.15%	-	0.71%	-0.03%	-	0.47%	0.21%	-	0.47%	0.17%
0.10	0.3	1.0	-	0.29%	-0.07%	-	0.32%	0.04%	-	0.32%	-0.02%	-	0.48%	-0.91%	-	0.43%	-0.02%	-	0.45%	-0.06%
0.15	0.3	1.0	-	0.18%	-0.36%	-	0.27%	-0.16%	-	0.22%	-0.30%	-	0.26%	-1.68%	-	0.34%	-0.29%	-	0.28%	-0.41%
0.20	0.3	1.0	-	0.33%	-0.40%	-	0.33%	-0.25%	-	0.09%	-0.62%	-	0.81%	-1.66%	-	0.40%	-0.36%	-	0.08%	-0.77%
0.25	0.3	1.0	-	0.17%	-0.71%	-	0.26%	-0.45%	-	0.28%	-0.59%	-	0.15%	-2.76%	-	0.34%	-0.56%	-	0.32%	-0.63%
0.30	0.3	1.0	-	0.33%	-0.74%	-	0.21%	-0.68%	-	0.31%	-0.73%	-	0.66%	-2.71%	-	0.19%	-0.84%	-	0.34%	-0.75%
0.35	0.3	1.0	-	0.27%	-0.95%	-	0.21%	-0.80%	-	0.21%	-1.00%	-	0.68%	-3.04%	-	0.26%	-0.83%	-	0.23%	-0.95%
0.40	0.3	1.0	-	0.41%	-0.96%	-	0.30%	-0.85%	-	0.18%	-1.19%	-	0.92%	-3.23%	-	0.34%	-0.82%	-	0.21%	-1.05%
0.45	0.3	1.0	-	0.16%	-1.40%	-	0.34%	-0.97%	-	0.19%	-1.33%	-	0.57%	-3.87%	-	0.36%	-0.88%	-	0.21%	-1.13%
0.50	0.3	1.0	-	0.11%	-1.55%	-	0.29%	-1.14%	-	0.20%	-1.47%	-	0.45%	-4.32%	-	0.33%	-1.00%	-	0.22%	-1.20%
0.3	**0.05**	1.0	-	0.06%	-0.20%	-	0.05%	-0.16%	-	0.09%	-0.17%	-	0.09%	-0.63%	-	0.05%	-0.16%	-	0.08%	-0.15%
0.3	**0.10**	1.0	-	0.16%	-0.32%	-	0.11%	-0.30%	-	0.11%	-0.38%	-	0.35%	-1.03%	-	0.11%	-0.31%	-	0.11%	-0.33%
0.3	**0.15**	1.0	-	0.15%	-0.52%	-	0.09%	-0.48%	-	0.11%	-0.58%	-	0.23%	-1.73%	-	0.08%	-0.51%	-	0.09%	-0.54%
0.3	**0.20**	1.0	-	0.19%	-0.64%	-	0.22%	-0.48%	-	0.20%	-0.65%	-	0.48%	-2.01%	-	0.24%	-0.49%	-	0.21%	-0.60%
0.3	**0.25**	1.0	-	0.39%	-0.55%	-	0.33%	-0.47%	-	0.27%	-0.70%	-	0.98%	-1.96%	-	0.38%	-0.51%	-	0.29%	-0.64%
0.3	**0.30**	1.0	-	0.33%	-0.74%	-	0.21%	-0.68%	-	0.31%	-0.73%	-	0.66%	-2.71%	-	0.19%	-0.84%	-	0.34%	-0.75%
0.3	**0.35**	1.0	-	0.24%	-0.91%	-	0.35%	-0.59%	-	0.24%	-0.88%	-	0.46%	-3.20%	-	0.40%	-0.70%	-	0.22%	-0.97%
0.3	**0.40**	1.0	-	0.47%	-0.73%	-	0.30%	-0.67%	-	0.15%	-1.00%	-	1.16%	-2.83%	-	0.39%	-0.82%	-	0.17%	-1.10%
0.3	**0.45**	1.0	-	0.36%	-0.87%	-	0.43%	-0.60%	-	0.46%	-0.75%	-	1.15%	-3.04%	-	0.52%	-0.76%	-	0.52%	-0.88%
0.3	**0.50**	1.0	-	0.47%	-0.84%	-	0.40%	-0.64%	-	0.39%	-0.84%	-	1.59%	-2.87%	-	0.54%	-0.84%	-	0.49%	-1.01%
0.3	0.3	**0.5**	-	0.47%	-0.10%	-	0.45%	-0.02%	-	0.47%	-0.09%	-	1.60%	-1.12%	-	0.46%	-0.09%	-	0.58%	-0.14%
0.3	0.3	**0.6**	-	0.46%	-0.27%	-	0.51%	-0.07%	-	0.40%	-0.32%	-	1.99%	-1.73%	-	0.68%	-0.15%	-	0.54%	-0.44%
0.3	0.3	**0.7**	-	0.47%	-0.39%	-	0.37%	-0.32%	-	0.45%	-0.38%	-	1.97%	-2.03%	-	0.56%	-0.44%	-	0.64%	-0.47%
0.3	0.3	**0.8**	-	0.29%	-0.66%	-	0.36%	-0.42%	-	0.25%	-0.67%	-	1.11%	-2.75%	-	0.50%	-0.53%	-	0.37%	-0.76%
0.3	0.3	**0.9**	-	0.34%	-0.68%	-	0.32%	-0.51%	-	0.33%	-0.65%	-	0.91%	-2.67%	-	0.39%	-0.62%	-	0.39%	-0.71%
0.3	0.3	**1.0**	-	0.33%	-0.74%	-	0.21%	-0.68%	-	0.31%	-0.73%	-	0.66%	-2.71%	-	0.19%	-0.84%	-	0.34%	-0.75%
0.3	0.3	**1.1**	-	0.15%	-0.94%	-	0.29%	-0.61%	-	0.11%	-0.98%	-	0.22%	-2.91%	-	0.31%	-0.67%	-	0.07%	-0.99%
0.3	0.3	**1.2**	-	0.32%	-0.78%	-	0.24%	-0.67%	-	-0.02%	-1.11%	-	0.23%	-2.70%	-	0.23%	-0.72%	-	-0.07%	-1.10%
0.3	0.3	**1.3**	-	0.09%	-1.03%	-	0.13%	-0.81%	-	0.24%	-0.87%	-	0.07%	-2.65%	-	0.08%	-0.85%	-	0.21%	-0.78%
0.3	0.3	**1.4**	-	0.21%	-0.91%	-	0.10%	-0.82%	-	0.01%	-1.09%	-	0.11%	-2.45%	-	0.03%	-0.86%	-	-0.05%	-1.02%
0.3	0.3	**1.5**	-	0.04%	-1.05%	-	0.08%	-0.84%	-	0.09%	-1.00%	-	-0.27%	-2.68%	-	0.06%	-0.80%	-	0.02%	-0.93%

In general, Model (CO) achieves higher fulfilment rates than Model (SAA) does, except in a few situations, while Model (MV) achieves lower fulfilment rates than Model (SAA) does. When we gradually increase ρ from 0.05 to 0.5, as shown in the first part of Table 8, the performance of Model (MV) gradually degrades, which implies the importance of considering correlations between nodes when correlations indeed exist. As c_v increases, as shown in the second part of Table 8, the performance gap between Model (CO) and Model (MV) also increases. This phenomenon indicates that when the variance of demand is sufficiently large, explicitly considering cross-moment information, instead of only marginal moment information, can further improve fulfilment rates.

We also summarize the relative changes in total cost (TC) compared to that of Model (SAA), as described in Table 9. The relative changes are calculated according to $\frac{TC_x - TC_{SAA}}{TC_{SAA}} \times 100\%, x \in \{CO, MV\}$. Similarly, Model (CO) dominates Model (SAA) by achieving lower costs in all but a few cases. Surprisingly, the maximum decrease in total cost can reach 14.96%.

Table 9: Total Cost Comparison of General Networks

Total Cost Relative Changes			MGD									MixD								
			(4, 8)			(8, 8)			(8, 12)			(4, 8)			(8, 8)			(8, 12)		
ρ	c_v	κ	SAA	CO	MV	SAA	CO	MV	SAA	CO	MV	SAA	CO	MV	SAA	CO	MV	SAA	CO	MV
0.05	0.3	1.0	-	-0.56%	-0.57%	-	-0.95%	-1.03%	-	-0.64%	-0.84%	-	-2.46%	0.30%	-	-1.69%	-1.01%	-	-1.66%	-0.94%
0.10	0.3	1.0	-	-0.92%	-0.68%	-	-0.48%	-0.48%	-	-0.74%	-0.69%	-	-1.75%	3.65%	-	-1.21%	-0.08%	-	-1.56%	-0.40%
0.15	0.3	1.0	-	-0.72%	-0.13%	-	-1.03%	-0.73%	-	-0.49%	-0.11%	-	-1.03%	6.80%	-	-1.47%	0.10%	-	-0.84%	0.64%
0.20	0.3	1.0	-	-0.86%	0.17%	-	-0.77%	-0.18%	-	-0.78%	0.13%	-	-3.26%	6.73%	-	-1.25%	0.54%	-	-0.69%	1.12%
0.25	0.3	1.0	-	-0.75%	0.66%	-	-0.68%	0.29%	-	-0.77%	0.52%	-	-0.55%	11.93%	-	-1.13%	1.02%	-	-1.01%	0.83%
0.30	0.3	1.0	-	-1.18%	0.86%	-	-1.14%	0.39%	-	-1.11%	0.71%	-	-2.93%	11.44%	-	-1.03%	1.42%	-	-1.27%	0.86%
0.35	0.3	1.0	-	-0.72%	1.71%	-	-0.80%	1.08%	-	-0.81%	1.54%	-	-2.99%	13.24%	-	-1.13%	1.25%	-	-0.91%	1.27%
0.40	0.3	1.0	-	-1.70%	1.21%	-	-0.90%	1.44%	-	-0.72%	2.23%	-	-4.53%	13.65%	-	-1.10%	1.28%	-	-0.91%	1.37%
0.45	0.3	1.0	-	-1.18%	2.46%	-	-1.18%	1.70%	-	-1.14%	2.32%	-	-3.54%	16.61%	-	-1.34%	1.13%	-	-1.25%	1.05%
0.50	0.3	1.0	-	-0.79%	3.09%	-	-1.13%	2.22%	-	-0.87%	3.14%	-	-2.84%	19.44%	-	-1.40%	1.22%	-	-1.03%	1.37%
0.3	**0.05**	1.0	-	-0.17%	0.50%	-	-0.16%	0.38%	-	-0.25%	0.48%	-	-0.38%	3.23%	-	-0.15%	0.38%	-	-0.21%	0.32%
0.3	**0.10**	1.0	-	-0.41%	0.77%	-	-0.35%	0.64%	-	-0.37%	0.94%	-	-1.56%	4.99%	-	-0.34%	0.69%	-	-0.33%	0.64%
0.3	**0.15**	1.0	-	-0.56%	0.99%	-	-0.45%	0.83%	-	-0.50%	1.15%	-	-1.01%	8.09%	-	-0.43%	1.09%	-	-0.42%	0.95%
0.3	**0.20**	1.0	-	-0.64%	1.15%	-	-0.61%	0.86%	-	-0.67%	1.23%	-	-2.18%	8.93%	-	-0.73%	0.96%	-	-0.73%	0.99%
0.3	**0.25**	1.0	-	-1.09%	0.71%	-	-0.95%	0.59%	-	-0.75%	1.23%	-	-4.19%	8.34%	-	-1.27%	0.84%	-	-0.88%	0.92%
0.3	**0.30**	1.0	-	-1.18%	0.86%	-	-1.14%	0.39%	-	-1.11%	0.71%	-	-2.93%	11.44%	-	-1.03%	1.42%	-	-1.27%	0.86%
0.3	**0.35**	1.0	-	-0.94%	1.11%	-	-0.93%	0.56%	-	-0.77%	1.02%	-	-2.12%	13.44%	-	-1.27%	1.22%	-	-0.71%	1.62%
0.3	**0.40**	1.0	-	-1.20%	0.69%	-	-0.73%	0.53%	-	-0.84%	0.75%	-	-4.63%	11.48%	-	-1.24%	1.47%	-	-0.98%	1.35%
0.3	**0.45**	1.0	-	-0.64%	0.99%	-	-1.22%	0.04%	-	-1.11%	0.38%	-	-4.65%	12.00%	-	-1.74%	1.04%	-	-1.44%	1.17%
0.3	**0.50**	1.0	-	-1.12%	0.61%	-	-0.89%	0.12%	-	-0.84%	0.48%	-	-6.45%	10.57%	-	-1.68%	1.37%	-	-1.37%	1.49%
0.3	0.3	**0.5**	-	-0.71%	-0.64%	-	-0.40%	-0.66%	-	-0.27%	-0.43%	-	-13.14%	10.46%	-	-0.24%	0.21%	-	-1.43%	0.32%
0.3	0.3	**0.6**	-	-1.15%	0.00%	-	-1.04%	-0.54%	-	-0.93%	0.13%	-	-14.96%	13.27%	-	-2.61%	0.30%	-	-2.34%	1.30%
0.3	0.3	**0.7**	-	-0.97%	0.86%	-	-0.70%	0.40%	-	-0.92%	0.47%	-	-12.62%	13.37%	-	-2.31%	1.45%	-	-2.62%	1.40%
0.3	0.3	**0.8**	-	-1.16%	0.84%	-	-0.88%	0.55%	-	-0.72%	1.00%	-	-6.75%	15.27%	-	-1.93%	1.41%	-	-1.65%	1.76%
0.3	0.3	**0.9**	-	-1.10%	0.91%	-	-0.91%	0.54%	-	-1.16%	0.64%	-	-4.47%	13.01%	-	-1.40%	1.34%	-	-1.51%	1.08%
0.3	0.3	**1.0**	-	-1.18%	0.86%	-	-1.14%	0.39%	-	-1.11%	0.71%	-	-2.93%	11.44%	-	-1.03%	1.42%	-	-1.27%	0.86%
0.3	0.3	**1.1**	-	-0.70%	1.19%	-	-0.82%	0.58%	-	-0.85%	0.96%	-	-0.94%	10.99%	-	-0.94%	0.96%	-	-0.66%	1.07%
0.3	0.3	**1.2**	-	-1.27%	0.46%	-	-0.71%	0.59%	-	-0.78%	0.90%	-	-0.84%	9.06%	-	-0.68%	0.85%	-	-0.55%	0.84%
0.3	0.3	**1.3**	-	-0.56%	1.05%	-	-0.59%	0.75%	-	-0.93%	0.64%	-	-0.45%	7.80%	-	-0.38%	0.94%	-	-0.81%	0.27%
0.3	0.3	**1.4**	-	-0.83%	0.66%	-	-0.59%	0.58%	-	-0.68%	0.77%	-	-0.38%	6.62%	-	-0.32%	0.78%	-	-0.40%	0.50%
0.3	0.3	**1.5**	-	-0.79%	0.49%	-	-0.49%	0.65%	-	-0.62%	0.70%	-	0.42%	6.46%	-	-0.41%	0.52%	-	-0.36%	0.42%
0.3	0.3	**1.4**	-	-0.83%	0.66%	-	-0.59%	0.58%	-	-0.68%	0.77%	-	-0.38%	6.62%	-	-0.32%	0.78%	-	-0.40%	0.50%
0.3	0.3	**1.5**	-	-0.79%	0.49%	-	-0.49%	0.65%	-	-0.62%	0.70%	-	0.42%	6.46%	-	-0.41%	0.52%	-	-0.36%	0.42%